

# T3 matched EFT Lagrangian terms: EFT\_1\_after\_F

EFT order: 5; loop order: 1. Rows are model configurations and columns are field configurations. This stage integrates out F. Remaining active T3 fields: S1, S2.

## Field combinations 1 of 6

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | $B_{\mu\nu}$                                | $D$                                                                        | $F$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | $H$ |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------------------|----------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{\hbar B_{\mu_1\mu_2}^2}{2\epsilon}$ | $+D_{\mu_1}\left(S_{2,j_1}\right)^\dagger D_{\mu_1}\left(S_{2,j_1}\right)$ | $\begin{aligned} & +\frac{2}{3}\hbar\log\left(\frac{\mu\hbar\alpha r^2}{M_F^2}\right)\frac{1}{\overline{W}_{2,1,\tau_1}W_{2,2,\tau_2}S_{2,j_2}^2}S_{2,j_1}^2W_{2,1,\tau_2}W_{2,2,\tau_1}\left(S_{2,j_1}\right)^2 \\ & -\frac{2\mathcal{C}N_{\text{wScalr}}22\hbar\log\left(\frac{\mu\hbar\alpha r^2}{M_F^2}\right)}{\sqrt{3}}\frac{1}{\overline{W}_{2,1,\tau_1}S_{2,j_1}W_{2,1,\tau_1}S_{2,j_1}^\dagger} \\ & -\frac{4}{3}\hbar\log\left(\frac{\mu\hbar\alpha r^2}{M_F^2}\right)\frac{1}{\overline{W}_{2,1,\tau_1}W_{2,2,\tau_2}S_{2,j_1}S_{2,j_2}W_{2,1,\tau_2}W_{2,2,\tau_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger} \\ & 2\hbar\log\left(\frac{\mu\hbar\alpha r^2}{M_F^2}\right)\frac{1}{\overline{W}_{2,1,\tau_1}S_{2,j_2}S_{2,j_4}W_{2,1,\tau_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger}I_{S2,j_3,j_1,j_4,j_2}^1\lambda_{S2}^1 \\ & -\frac{2\hbar\log\left(\frac{\mu\hbar\alpha r^2}{M_F^2}\right)}{\sqrt{3}}\frac{1}{\overline{W}_{2,1,\tau_1}S_{2,j_3}S_{2,j_4}W_{2,1,\tau_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger}I_{S2,j_3,j_1,j_4,j_2}^2\lambda_{S2}^2 \end{aligned}$ |     |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              |                                             | $+D_{\mu_1}\left(S_{2,j_1}\right)^\dagger D_{\mu_1}\left(S_{2,j_1}\right)$ | $\begin{aligned} & +\frac{2}{3}\hbar\log\left(\frac{\mu\hbar\alpha r^2}{M_F^2}\right)\frac{1}{\overline{W}_{2,1,\tau_1}W_{2,2,\tau_2}S_{2,j_1}S_{2,j_2}W_{2,1,\tau_2}W_{2,2,\tau_1}\left(S_{2,j_1}\right)^2} \\ & -\frac{2\mathcal{C}N_{\text{wScalr}}22\hbar\log\left(\frac{\mu\hbar\alpha r^2}{M_F^2}\right)}{\sqrt{3}}\frac{1}{\overline{W}_{2,1,\tau_1}S_{2,j_1}W_{2,1,\tau_1}S_{2,j_1}^\dagger} \\ & -\frac{4}{3}\hbar\log\left(\frac{\mu\hbar\alpha r^2}{M_F^2}\right)\frac{1}{\overline{W}_{2,1,\tau_1}W_{2,2,\tau_2}S_{2,j_1}S_{2,j_2}W_{2,1,\tau_2}W_{2,2,\tau_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger} \\ & 2\hbar\log\left(\frac{\mu\hbar\alpha r^2}{M_F^2}\right)\frac{1}{\overline{W}_{2,1,\tau_1}S_{2,j_2}S_{2,j_4}W_{2,1,\tau_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger}I_{S2,j_3,j_1,j_4,j_2}^1\lambda_{S2}^1 \\ & -\frac{2\hbar\log\left(\frac{\mu\hbar\alpha r^2}{M_F^2}\right)}{\sqrt{3}}\frac{1}{\overline{W}_{2,1,\tau_1}S_{2,j_3}S_{2,j_4}W_{2,1,\tau_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger}I_{S2,j_3,j_1,j_4,j_2}^2\lambda_{S2}^2 \end{aligned}$     |     |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{\hbar B_{\mu_1\mu_2}^2}{2\epsilon}$ | $+D_{\mu_1}\left(S_{2,j_1}\right)^\dagger D_{\mu_1}\left(S_{2,j_1}\right)$ | $\begin{aligned} & +\frac{2}{3}\hbar\log\left(\frac{\mu\hbar\alpha r^2}{M_F^2}\right)\frac{1}{\overline{W}_{2,1,\tau_1}W_{2,2,\tau_2}S_{2,j_1}S_{2,j_2}W_{2,1,\tau_2}W_{2,2,\tau_1}\left(S_{2,j_1}\right)^2} \\ & -\frac{2\mathcal{C}N_{\text{wScalr}}22\hbar\log\left(\frac{\mu\hbar\alpha r^2}{M_F^2}\right)}{\sqrt{3}}\frac{1}{\overline{W}_{2,1,\tau_1}S_{2,j_1}W_{2,1,\tau_1}S_{2,j_1}^\dagger} \\ & -\frac{4}{3}\hbar\log\left(\frac{\mu\hbar\alpha r^2}{M_F^2}\right)\frac{1}{\overline{W}_{2,1,\tau_1}W_{2,2,\tau_2}S_{2,j_1}S_{2,j_2}W_{2,1,\tau_2}W_{2,2,\tau_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger} \\ & 2\hbar\log\left(\frac{\mu\hbar\alpha r^2}{M_F^2}\right)\frac{1}{\overline{W}_{2,1,\tau_1}S_{2,j_2}S_{2,j_4}W_{2,1,\tau_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger}I_{S2,j_3,j_1,j_4,j_2}^1\lambda_{S2}^1 \\ & -\frac{2\hbar\log\left(\frac{\mu\hbar\alpha r^2}{M_F^2}\right)}{\sqrt{3}}\frac{1}{\overline{W}_{2,1,\tau_1}S_{2,j_3}S_{2,j_4}W_{2,1,\tau_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger}I_{S2,j_3,j_1,j_4,j_2}^2\lambda_{S2}^2 \end{aligned}$     |     |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{2\hbar B_{\mu_1\mu_2}^2}{\epsilon}$ | $+D_{\mu_1}\left(S_{2,j_1}\right)^\dagger D_{\mu_1}\left(S_{2,j_1}\right)$ | $\begin{aligned} & +\frac{2}{3}\hbar\log\left(\frac{\mu\hbar\alpha r^2}{M_F^2}\right)\frac{1}{\overline{W}_{2,1,\tau_1}W_{2,2,\tau_2}S_{2,j_1}S_{2,j_2}W_{2,1,\tau_2}W_{2,2,\tau_1}\left(S_{2,j_1}\right)^2} \\ & -\frac{2\mathcal{C}N_{\text{wScalr}}22\hbar\log\left(\frac{\mu\hbar\alpha r^2}{M_F^2}\right)}{\sqrt{3}}\frac{1}{\overline{W}_{2,1,\tau_1}S_{2,j_1}W_{2,1,\tau_1}S_{2,j_1}^\dagger} \\ & -\frac{4}{3}\hbar\log\left(\frac{\mu\hbar\alpha r^2}{M_F^2}\right)\frac{1}{\overline{W}_{2,1,\tau_1}W_{2,2,\tau_2}S_{2,j_1}S_{2,j_2}W_{2,1,\tau_2}W_{2,2,\tau_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger} \\ & 2\hbar\log\left(\frac{\mu\hbar\alpha r^2}{M_F^2}\right)\frac{1}{\overline{W}_{2,1,\tau_1}S_{2,j_2}S_{2,j_4}W_{2,1,\tau_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger}I_{S2,j_3,j_1,j_4,j_2}^1\lambda_{S2}^1 \\ & -\frac{2\hbar\log\left(\frac{\mu\hbar\alpha r^2}{M_F^2}\right)}{\sqrt{3}}\frac{1}{\overline{W}_{2,1,\tau_1}S_{2,j_3}S_{2,j_4}W_{2,1,\tau_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger}I_{S2,j_3,j_1,j_4,j_2}^2\lambda_{S2}^2 \end{aligned}$     |     |

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| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | $B\mu\nu$                                  | $D$                                                  | $F$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | $H$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{\hbar B^2 \mu_1 \mu_2}{2\epsilon}$ | $+D_{\mu_1}(S_{1,j_1})^\dagger D_{\mu_1}(S_{1,j_1})$ | $\begin{aligned} & +\frac{2}{3}\hbar\log\left(\frac{\mu_B\kappa^2}{M_F^2}\right)\frac{1}{\overline{W}_{1,\tau_1,\tau_1}\overline{W}_{1,\tau_2,\tau_2}}S_{1,j_2}^2\overline{W}_{1,\tau_1,\tau_2}\overline{W}_{1,\tau_2,\tau_1}\left(S_{1,j_1}\right)^2 \\ & -\frac{\Im C\text{NewScalar}12\hbar\log\left(\frac{\mu_B\lambda^2}{M_F^2}\right)}{\sqrt{3}}\frac{1}{\overline{W}_{1,\tau_1,\tau_1}S_{1,j_1}}\overline{W}_{1,\tau_1,\tau_1}S_{1,j_1}^\dagger \\ & -\frac{4}{3}\hbar\log\left(\frac{\mu_B\kappa^2}{M_F^2}\right)\frac{1}{\overline{W}_{1,\tau_1,\tau_1}\overline{W}_{1,\tau_2,\tau_2}}S_{1,j_1}S_{1,j_2}\overline{W}_{1,\tau_1,\tau_2}\overline{W}_{1,\tau_2,\tau_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger \\ & -\frac{2\hbar\log\left(\frac{\mu_B\lambda^2}{M_F^2}\right)}{\sqrt{3}}\frac{1}{\overline{W}_{1,\tau_1,\tau_1}S_{1,j_1}S_{1,j_2}}\overline{W}_{1,\tau_1,\tau_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger\mathcal{I}_{S_{1,j_1},j_1;j_2,j_2}{}^1\lambda_{S_1}^1 \\ & -\frac{2\hbar\log\left(\frac{\mu_B\lambda^2}{M_F^2}\right)}{\sqrt{3}}\frac{1}{\overline{W}_{1,\tau_1,\tau_1}S_{1,j_1}S_{1,j_2}}\overline{W}_{1,\tau_1,\tau_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger\mathcal{I}_{S_{1,j_1},j_1;j_2,j_2}{}^2\lambda_{S_1}^2 \end{aligned}$ | $\begin{aligned} & +\frac{2\overline{W}_{1,\tau_1,\tau_1}\overline{W}_{1,\tau_2,\tau_2}L_{\kappa,\kappa,\varphi}^{-T}C P_\kappa L_{\kappa,\varphi,j_1}H_{\kappa_1}H_{\kappa_2}\lambda_{\varphi j_1}\text{Bar}\left(\exp\left(S U(2)_1\right)\right)_{\kappa_1,\kappa_2}\text{Bar}\left(\exp\left(S U(2)_1\right)\right)_{\kappa_1,\kappa_2}}{\sqrt{\delta} M_F} \\ & +\frac{2\overline{W}_{1,\tau_1,\tau_1}\overline{W}_{1,\tau_2,\tau_2}L_{\kappa,\kappa,\varphi}^{-T}C P_\kappa L_{\kappa,\varphi,j_1}H_{\kappa_1}H_{\kappa_2}\lambda_{\varphi j_1}\text{Bar}\left(\exp\left(S U(2)_1\right)\right)_{\kappa_1,\kappa_2}\text{Bar}\left(\exp\left(S U(2)_1\right)\right)_{\kappa_1,\kappa_2}}{\sqrt{\delta} M_F} \\ & +\frac{2\overline{W}_{1,\tau_1,\tau_1}C P_\kappa L_{\kappa,\kappa,\varphi}^{-T}P_\kappa L_{\kappa,\kappa,\varphi}^{-T}E_{\varphi j_1,j_1}\overline{W}_{1,\tau_1,\tau_1}\overline{W}_{1,\tau_2,\tau_2}\text{CPH}\left(S U(2)_1\right)_{\kappa_1,\kappa_2}\text{QH}\left(S U(2)_1\right)_{\kappa_1,\kappa_2}H_{\kappa_1}^\dagger H_{\kappa_2}^\dagger}{\sqrt{\delta} M_F} \\ & +\frac{2\overline{W}_{1,\tau_1,\tau_1}C P_\kappa L_{\kappa,\kappa,\varphi}^{-T}P_\kappa L_{\kappa,\kappa,\varphi}^{-T}E_{\varphi j_1,j_1}\overline{W}_{1,\tau_1,\tau_1}\overline{W}_{1,\tau_2,\tau_2}\text{CPH}\left(S U(2)_1\right)_{\kappa_1,\kappa_2}\text{QH}\left(S U(2)_1\right)_{\kappa_1,\kappa_2}H_{\kappa_1}^\dagger H_{\kappa_2}^\dagger}{\sqrt{\delta} M_F} \\ & -\frac{2\overline{W}_{1,\tau_1,\tau_1}Y_{\kappa,\kappa,j_1}H_{\kappa_1}S_{1,j_1}\overline{W}_{1,\tau_1,\tau_1}Y_{\kappa,\kappa,j_1}H_{\kappa_2}^\dagger S_{1,j_1}^\dagger}{\sqrt{\delta}} \\ & -\frac{2\overline{W}_{1,\tau_1,\tau_1}Y_{\kappa,\kappa,j_1}H_{\kappa_1}S_{1,j_1}\overline{W}_{1,\tau_1,\tau_1}Y_{\kappa,\kappa,j_1}H_{\kappa_2}^\dagger S_{1,j_1}^\dagger}{\sqrt{\delta}} \\ & -\frac{2i\sqrt{\frac{2}{3}}\overline{W}_{1,\tau_1,\tau_1}Y_{\kappa,\kappa,j_1}H_{\kappa_2}S_{1,j_2}\overline{W}_{1,\tau_2,\tau_2}Y_{\kappa,\kappa,j_1}H_{\kappa_1}^\dagger S_{1,j_1}^\dagger\mathcal{I}_{\text{Tm }1,\kappa_1;\kappa_2,j_2,j_1}{}^1}{\epsilon} \\ & -2i\sqrt{\frac{2}{3}}\overline{W}_{1,\tau_1,\tau_1}Y_{\kappa,\kappa,j_1}H_{\kappa_2}S_{1,j_2}\overline{W}_{1,\tau_2,\tau_2}Y_{\kappa,\kappa,j_1}H_{\kappa_1}^\dagger S_{1,j_1}^\dagger\mathcal{I}_{\text{Tm }1,\kappa_1;\kappa_2,j_2,j_1}{}^1 \\ & +H_{\kappa_2}S_{1,j_2}H_{\kappa_1}^\dagger S_{1,j_1}^\dagger\mathcal{I}_{\text{Tm }1,\kappa_1;\kappa_2,j_2,j_1}{}^2\lambda_{\text{Tm}}^1 \\ & -\frac{2\overline{W}_{1,\tau_1,\tau_1}H_{\kappa_2}S_{1,j_2}\overline{W}_{1,\tau_1,\tau_1}H_{\kappa_1}^\dagger S_{1,j_1}^\dagger\mathcal{I}_{\text{Tm }1,\kappa_1;\kappa_2,j_2,j_1}{}^2\lambda_{\text{Tm}}^1}{\sqrt{\delta}} \\ & -\frac{\Delta\overline{W}_{1,\tau_1,\tau_1}H_{\kappa_2}S_{1,j_2}\overline{W}_{1,\tau_1,\tau_1}H_{\kappa_1}^\dagger S_{1,j_1}^\dagger\mathcal{I}_{\text{Tm }1,\kappa_1;\kappa_2,j_2,j_1}{}^2\lambda_{\text{Tm}}^2}{\sqrt{\delta}} \end{aligned}$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              |                                            | $+D_{\mu_1}(S_{1,j_1})^\dagger D_{\mu_1}(S_{1,j_1})$ | $\begin{aligned} & +\frac{2}{3}\hbar\log\left(\frac{\mu_B\kappa^2}{M_F^2}\right)\frac{1}{\overline{W}_{1,\tau_1,\tau_1}\overline{W}_{1,\tau_2,\tau_2}}S_{1,j_2}^2\overline{W}_{1,\tau_1,\tau_2}\overline{W}_{1,\tau_2,\tau_1}\left(S_{1,j_1}\right)^2 \\ & -\frac{\Im C\text{NewScalar}12\hbar\log\left(\frac{\mu_B\lambda^2}{M_F^2}\right)}{\sqrt{3}}\frac{1}{\overline{W}_{1,\tau_1,\tau_1}S_{1,j_1}}\overline{W}_{1,\tau_1,\tau_1}S_{1,j_1}^\dagger \\ & -\frac{4}{3}\hbar\log\left(\frac{\mu_B\kappa^2}{M_F^2}\right)\frac{1}{\overline{W}_{1,\tau_1,\tau_1}\overline{W}_{1,\tau_2,\tau_2}}S_{1,j_1}S_{1,j_2}\overline{W}_{1,\tau_1,\tau_2}\overline{W}_{1,\tau_2,\tau_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger \\ & -\frac{2\hbar\log\left(\frac{\mu_B\lambda^2}{M_F^2}\right)}{\sqrt{3}}\frac{1}{\overline{W}_{1,\tau_1,\tau_1}S_{1,j_1}S_{1,j_2}}\overline{W}_{1,\tau_1,\tau_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger\mathcal{I}_{S_{1,j_1},j_1;j_2,j_2}{}^1\lambda_{S_1}^1 \\ & -\frac{2\hbar\log\left(\frac{\mu_B\lambda^2}{M_F^2}\right)}{\sqrt{3}}\frac{1}{\overline{W}_{1,\tau_1,\tau_1}S_{1,j_1}S_{1,j_2}}\overline{W}_{1,\tau_1,\tau_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger\mathcal{I}_{S_{1,j_1},j_1;j_2,j_2}{}^2\lambda_{S_1}^2 \end{aligned}$ | $\begin{aligned} & +\frac{2\overline{W}_{1,\tau_1,\tau_1}\overline{W}_{1,\tau_2,\tau_2}L_{\kappa,\kappa,\varphi}^{-T}C P_\kappa L_{\kappa,\varphi,j_1}H_{\kappa_1}H_{\kappa_2}\lambda_{\varphi j_1}\text{Bar}\left(\exp\left(S U(2)_1\right)\right)_{\kappa_1,\kappa_2}\text{Bar}\left(\exp\left(S U(2)_1\right)\right)_{\kappa_1,\kappa_2}}{\sqrt{\delta} M_F} \\ & +\frac{2\overline{W}_{1,\tau_1,\tau_1}\overline{W}_{1,\tau_2,\tau_2}L_{\kappa,\kappa,\varphi}^{-T}C P_\kappa L_{\kappa,\varphi,j_1}H_{\kappa_1}H_{\kappa_2}\lambda_{\varphi j_1}\text{Bar}\left(\exp\left(S U(2)_1\right)\right)_{\kappa_1,\kappa_2}\text{Bar}\left(\exp\left(S U(2)_1\right)\right)_{\kappa_1,\kappa_2}}{\sqrt{\delta} M_F} \\ & +\frac{2\overline{W}_{1,\tau_1,\tau_1}C P_\kappa L_{\kappa,\kappa,\varphi}^{-T}P_\kappa L_{\kappa,\kappa,\varphi}^{-T}E_{\varphi j_1,j_1}\overline{W}_{1,\tau_1,\tau_1}\overline{W}_{1,\tau_2,\tau_2}\text{CPH}\left(S U(2)_1\right)_{\kappa_1,\kappa_2}\text{QH}\left(S U(2)_1\right)_{\kappa_1,\kappa_2}H_{\kappa_1}^\dagger H_{\kappa_2}^\dagger}{\sqrt{\delta} M_F} \\ & +\frac{2\overline{W}_{1,\tau_1,\tau_1}C P_\kappa L_{\kappa,\kappa,\varphi}^{-T}P_\kappa L_{\kappa,\kappa,\varphi}^{-T}E_{\varphi j_1,j_1}\overline{W}_{1,\tau_1,\tau_1}\overline{W}_{1,\tau_2,\tau_2}\text{CPH}\left(S U(2)_1\right)_{\kappa_1,\kappa_2}\text{QH}\left(S U(2)_1\right)_{\kappa_1,\kappa_2}H_{\kappa_1}^\dagger H_{\kappa_2}^\dagger}{\sqrt{\delta} M_F} \\ & -\frac{2\overline{W}_{1,\tau_1,\tau_1}Y_{\kappa,\kappa,j_1}H_{\kappa_1}S$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

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| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | $B_{\mu\nu}$ | $D$ | $F$ | $H$ |
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## Field combinations 2 of 6

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| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$ | $S_1$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | $S_2$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | $W_{\mu\nu}$                                           | constant |
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|       |          |           |               |           |               |       |       | $\begin{aligned} & -\frac{24M_{Pl,1,r_1}^2 W_{1,2,r_2} S_{1,2,r_1}^2 Y_{1,1,r_1} Y_{2,1,r_2} Y_{1,2,r_1} \left(S_1^\dagger\right)^2}{e} \\ & + CNewScalar12S_1 S_1^\dagger \\ & - CNewScalar12M_{Pl,1,r_1}^2 S_{1,2,r_1} Y_{1,1,r_1} S_1^\dagger \\ & + \frac{2CNewScalar12M_{Pl,1,r_1}^2 S_{1,2,r_1} Y_{1,1,r_1} S_1^\dagger}{e} \\ & - \frac{90M_{Pl,1,r_1}^2 W_{1,2,r_2} L_{a,b,c}}{4eM_F} T C P_k L_{a_1,1,1} g_{S_2,1}^2 Z_{T1,a_1,2,g,1} S_1^\dagger \\ & - \frac{360M_{Pl,1,r_1}^2 W_{1,2,r_2} L_{a,b,c}}{8M_F} T C P_k L_{a_1,1,1} g_{S_2,1}^2 Z_{T1,a_1,2,g,1} S_1^\dagger \\ & + \frac{M_{Pl,1,r_1}^2 W_{1,2,r_2} L_{a,b,c}}{M_F} T C P_k L_{a_1,1,1} S_{2,j,1} Z_{T1,a_1,2,g,1} S_1^\dagger \\ & - \frac{M_{Pl,1,r_1}^2 W_{1,1,r_2} W_{2,1,r_2} L_{a,b,c}}{4eM_F} T C P_k L_{a_1,1,1} S_{2,j,1} W_{1,1,r_2} Z_{T1,a_1,2,g,1} S_1^\dagger \\ & - \frac{360M_{Pl,1,r_1}^2 W_{1,1,r_2} W_{2,1,r_2} L_{a,b,c}}{8M_F} T C P_k L_{a_1,1,1} S_{2,j,1} W_{1,1,r_2} Z_{T1,a_1,2,g,1} S_1^\dagger \\ & - \frac{M_{Pl,1,r_1}^2 W_{1,1,r_2} W_{2,1,r_2} L_{a,b,c}}{cM_F} T C P_k L_{a_1,1,1} S_{2,j,1} W_{1,1,r_2} Z_{T1,a_1,2,g,1} S_1^\dagger \\ & - \frac{M_{Pl,1,r_1}^2 W_{1,1,r_2} W_{2,1,r_2} L_{a,b,c}}{2M_F} T C P_k L_{a_1,1,1} S_{2,j,1} W_{1,1,r_2} Z_{T1,a_1,2,g,1} S_1^\dagger \\ & - \frac{M_{Pl,1,r_1}^2 W_{1,1,r_2} W_{2,1,r_2} L_{a,b,c}}{4eM_F} T C P_k L_{a_1,1,1} S_{2,j,1} W_{1,1,r_2} Z_{T1,a_1,2,g,1} S_1^\dagger \\ & - \frac{360M_{Pl,1,r_1}^2 W_{1,1,r_2} W_{2,1,r_2} L_{a,b,c}}{8M_F} T C P_k L_{a_1,1,1} S_{2,j,1} W_{1,1,r_2} Z_{T1,a_1,2,g,1} S_1^\dagger \\ & - \sqrt{3} \frac{M_{Pl,1,r_1}^2 W_{1,1,r_2} W_{2,1,r_2} L_{a,b,c}}{4eM_F} T C P_k L_{a_1,1,1} S_{2,j,1} W_{1,1,r_2} Z_{T1,a_1,2,g,1} S_1^\dagger \\ & - \frac{3\sqrt{3} M_{Pl,1,r_1}^2 W_{1,1,r_2} W_{2,1,r_2} L_{a,b,c}}{4eM_F} T C P_k L_{a_1,1,1} S_{2,j,1} W_{1,1,r_2} Z_{T1,a_1,2,g,1} S_1^\dagger \\ & - \sqrt{3} \frac{M_{Pl,1,r_1}^2 W_{1,1,r_2} W_{2,1,r_2} L_{a,b,c}}{4eM_F} T C P_k L_{a_1,1,1} S_{2,j,1} W_{1,1,r_2} Z_{T1,a_1,2,g,1} S_1^\dagger \\ & - \frac{3\sqrt{3} M_{Pl,1,r_1}^2 W_{1,1,r_2} W_{2,1,r_2} L_{a,b,c}}{4eM_F} T C P_k L_{a_1,1,1} S_{2,j,1} W_{1,1,r_2} Z_{T1,a_1,2,g,1} S_1^\dagger \\ & - \frac{M_{Pl,1,r_1}^2 W_{1,1,r_2} W_{2,1,r_2} L_{a,b,c}}{\sqrt{3} eM_F} T C P_k L_{a_1,1,1} S_{2,j,1} W_{1,1,r_2} Z_{T1,a_1,2,g,1} S_1^\dagger \\ & - \frac{M_{Pl,1,r_1}^2 W_{1,1,r_2} W_{2,1,r_2} L_{a,b,c}}{2\sqrt{3} M_F} T C P_k L_{a_1,1,1} S_{2,j,1} W_{1,1,r_2} Z_{T1,a_1,2,g,1} S_1^\dagger \\ & - \frac{M_{Pl,1,r_1}^2 L_{a,b,c}}{C P_k L_{a,b,c}} S_1 W_{1,1,r_2} W_{2,1,r_2} W_{1,1,r_2} \text{Bar(T3MI acG)}_{1,1,2,g,1} S_{2,j,1}^\dagger \\ & - \frac{360M_{Pl,1,r_1}^2 L_{a,b,c}}{8M_F} C P_k L_{a,b,c} S_1 W_{1,1,r_2} W_{1,1,r_2} W_{2,1,r_2} \text{Bar(T3MI acG)}_{1,1,2,g,1} S_{2,j,1}^\dagger \\ & - \frac{90M_{Pl,1,r_1}^2 C P_k L_{a,b,c}}{4eM_F} g_{S_1}^2 S_{1,1,r_1} W_{2,1,r_2} \text{Bar(T3MI acG)}_{1,1,2,g,1} S_{2,j,1}^\dagger \\ & - \frac{36M_{Pl,1,r_1}^2 C P_k L_{a,b,c}}{4M_F} g_{S_1}^2 S_{1,1,r_1} W_{1,1,r_2} W_{2,1,r_2} \text{Bar(T3MI acG)}_{1,1,2,g,1} S_{2,j,1}^\dagger \\ & + \frac{M_{Pl,1,r_1}^2 C P_k L_{a,b,c}}{M_F} S_1 W_{1,1,r_1} W_{2,1,r_1} \text{Bar(T3MI acG)}_{1,1,2,g,1} S_{2,j,1}^\dagger \\ & - \sqrt{3} \frac{M_{Pl,1,r_1}^2 L_{a,b,c}}{C P_k L_{a,b,c}} S_1 W_{1,1,r_2} W_{2,1,r_2} W_{1,1,r_2} \text{Bar(T3MI acG)}_{1,1,2,g,1} S_{2,j,1}^\dagger \\ & - \frac{3\sqrt{3} M_{Pl,1,r_1}^2 L_{a,b,c}}{4eM_F} C P_k L_{a,b,c} S_1 W_{1,1,r_2} W_{2,1,r_2} W_{1,1,r_2} \text{Bar(T3MI acG)}_{1,1,2,g,1} S_{2,j,1}^\dagger \\ & - \frac{M_{Pl,1,r_1}^2 L_{a,b,c}}{C P_k L_{a,b,c}} S_1 W_{1,1,r_2} W_{2,1,r_2} W_{1,1,r_2} \text{Bar(T3MI acG)}_{1,1,2,g,1} S_{2,j,1}^\dagger \\ & - \frac{M_{Pl,1,r_1}^2 L_{a,b,c}}{2M_F} C P_k L_{a,b,c} S_1 W_{1,1,r_2} W_{2,1,r_2} W_{1,1,r_2} \text{Bar(T3MI acG)}_{1,1,2,g,1} S_{2,j,1}^\dagger \\ & - \frac{M_{Pl,1,r_1}^2 L_{a,b,c}}{\sqrt{3} eM_F} C P_k L_{a,b,c} S_1 W_{1,1,r_2} W_{2,1,r_2} W_{1,1,r_2} \text{Bar(T3MI acG)}_{1,1,2,g,1} S_{2,j,1}^\dagger \\ & - \frac{M_{Pl,1,r_1}^2 L_{a,b,c}}{2\sqrt{3} M_F} C P_k L_{a,b,c} S_1 W_{1,1,r_2} W_{2,1,r_2} W_{1,1,r_2} \text{Bar(T3MI acG)}_{1,1,2,g,1} S_{2,j,1}^\dagger \\ & - \frac{M_{Pl,1,r_1}^2 L_{a,b,c}}{4eM_F} C P_k L_{a,b,c} S_1 W_{1,1,r_2} W_{2,1,r_2} W_{1,1,r_2} \text{Bar(T3MI acG)}_{1,1,2,g,1} S_{2,j,1}^\dagger \\ & - \frac{3\sqrt{3} M_{Pl,1,r_1}^2 L_{a,b,c}}{4M_F} C P_k L_{a,b,c} S_1 W_{1,1,r_2} W_{2,1,r_2} W_{1,1,r_2} \text{Bar(T3MI acG)}_{1,1,2,g,1} S_{2,j,1}^\dagger \\ & - \frac{3\sqrt{3} M_{Pl,1,r_1}^2 L_{a,b,c}}{4M_F} C P_k L_{a,b,c} S_1 W_{1,1,r_2} W_{2,1,r_2} W_{1,1,r_2} \text{Bar(T3MI acG)}_{1,1,2,g,1} S_{2,j,1}^\dagger \\ & - \frac{48M_{Pl,1,r_1}^2 W_{1,1,r_2} S_1 S_{2,1,r_1} W_{2,1,r_2} S_1^\dagger S_{2,j,1}^\dagger}{\sqrt{3} e} \\ & + \frac{48M_{Pl,1,r_1}^2 W_{1,1,r_2} S_1 S_{2,1,r_1} W_{1,1,r_2} W_{2,1,r_2} S_1^\dagger S_{2,j,1}^\dagger}{\sqrt{3}} \\ & + \frac{48M_{Pl,1,r_1}^2 W_{1,1,r_2} S_1 S_{2,1,r_1} W_{2,1,r_2} W_{2,1,r_2} S_1^\dagger S_{2,j,1}^\dagger}{\sqrt{3}} \\ & - \frac{M_{Pl,1,r_1}^2 W_{1,1,r_2} W_{2,1,r_2} L_{a,b,c}}{eM_F} T C P_k L_{a_1,1,1} S_{2,j,1} Z_{T1,a_1,2,g,1} S_1^\dagger \lambda_{12}^1 \\ & - \frac{M_{Pl,1,r_1}^2 W_{1,1,r_2} W_{2,1,r_2} L_{a,b,c}}{M_F} T C P_k L_{a_1,1,1} S_{2,j,1} Z_{T1,a_1,2,g,1} S_1^\dagger \lambda_{12}^2 \\ & - \frac{M_{Pl,1,r_1}^2 C P_k L_{a,b,c}}{eM_F} S_1 W_{1,1,r_1} W_{2,1,r_2} \text{Bar(T3MI acG)}_{1,1,2,g,1} S_{2,j,1}^\dagger \lambda_{12}^1 \\ & - \frac{M_{Pl,1,r_1}^2 C P_k L_{a,b,c}}{M_F} S_1 W_{1,1,r_1} W_{2,1,r_2} \text{Bar(T3MI acG)}_{1,1,2,g,1} S_{2,j,1}^\dagger \lambda_{12}^2 \\ & + S_1 S_{2,1,r_2} S_1^\dagger S_{2,1,r_1}^\dagger I_{12,1,2-j_1}^1 \lambda_{12}^1 \\ & - \frac{2M_{Pl,1,r_1}^2 S_1 S_{2,1,r_2} W_{1,1,r_1} S_1^\dagger S_{2,1,r_1}^\dagger I_{12,1,2-j_1}^1 \lambda_{12}^1}{e} \\ & - \frac{M_{Pl,1,r_1}^2 S_1 S_{2,1,r_2} W_{1,1,r_1} S_1^\dagger S_{2,1,r_1}^\dagger I_{12,1,2-j_1}^1 \lambda_{12}^1}{e} \\ & - \frac{2M_{Pl,1,r_1}^2 S_1 S_{2,1,r_2} W_{1,1,r_1} S_1^\dagger S_{2,1,r_1}^\dagger I_{12,1,2-j_1}^1 \lambda_{12}^1}{e} \\ & - \frac{M_{Pl,1,r_1}^2 S_1 S_{2,1,r_2} W_{1,1,r_1} S_1^\dagger S_{2,1,r_1}^\dagger I_{12,1,2-j_1}^1 \lambda_{12}^1}{\sqrt{3} e} \\ & + \frac{1}{2} g_{S_1}^2 \left(S_1^\dagger\right)^2 \lambda_{11}^1 \\ & - \frac{2M_{Pl,1,r_1}^2 g_{S_1}^2 W_{1,1,r_1} S_1^\dagger \left(S_1^\dagger\right)^2 \lambda_{11}^1}{e} \\ & - \frac{M_{Pl,1,r_1}^2 g_{S_1}^2 W_{1,1,r_1} S_1^\dagger \left(S_1^\dagger\right)^2 \lambda_{11}^1}{e} \end{aligned}$ | $\begin{aligned} & + \frac{2h\overline{y}_{2,i_1,r_1} \overline{y}_{2,i_2,r_2} S_{2,j_2}^2 \overline{y}_{2,i_1,r_2} \overline{y}_{2,i_2,r_1} \left(S_{2,j_1}^\dagger\right)^2}{3e} \\ & + CNewScalar22S_{2,j_1} S_{2,j_1}^\dagger \\ & - \frac{CNewScalar22h\overline{y}_{2,i_1,r_1} S_{2,j_1} \overline{y}_{2,i_1,r_1} S_{2,j_1}^\dagger}{\sqrt{3}} \\ & - \frac{2CNewScalar22h\overline{y}_{2,i_1,r_1} S_{2,j_1} \overline{y}_{2,i_1,r_1} S_{2,j_1}^\dagger}{\sqrt{3} e} \\ & - \frac{4h\overline{y}_{2,i_1,r_1} \overline{y}_{2,i_2,r_2} S_{2,j_1} S_{2,j_2} \overline{y}_{2,i_1,r_2} \overline{y}_{2,i_2,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger}{3e} \\ & + \frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger I_{S2,j_3,j_1,j_4,j_2}^1 \lambda_{S2}^1 \\ & - \frac{2h\overline{y}_{2,i_1,r_1} S_{2,j_3} S_{2,j_4} \overline{y}_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger I_{S2,j_3,j_1,j_4,j_2}^1 \lambda_{S2}^1}{\sqrt{3} e} \\ & - \frac{h\overline{y}_{2,i_1,r_1} S_{2,j_3} S_{2,j_4} \overline{y}_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger I_{S2,j_3,j_1,j_4,j_2}^1 \lambda_{S2}^1}{\sqrt{3}} \\ & + \frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger I_{S2,j_3,j_1,j_4,j_2}^2 \lambda_{S2}^2 \\ & - \frac{2h\overline{y}_{2,i_1,r_1} S_{2,j_3} S_{2,j_4} \overline{y}_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger I_{S2,j_3,j_1,j_4,j_2}^2 \lambda_{S2}^2}{\sqrt{3} e} \\ & - \frac{h\overline{y}_{2,i_1,r_1} S_{2,j_3} S_{2,j_4} \overline{y}_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger I_{S2,j_3,j_1,j_4,j_2}^2 \lambda_{S2}^2}{\sqrt{3}} \end{aligned}$ |                                                        |          |
| T3-A  | 1        | 1         | $\frac{1}{2}$ | 3         | $\frac{3}{2}$ | 2     | 1     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             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                                                   | $-\frac{\hbar\left(W_{\mu_1\mu_2}^{A_1}\right)^2}{2e}$ |          |

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$ | $S_1$ |
|-------|----------|-----------|----------------|-----------|---------------|-------|-------|-------|
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0     |       |

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | $S_1$ | $S_2$ | $W_{\mu\nu}$ | constant |
|-------|----------|-----------|-----------|-----------|-----------|-------|-------|-------|-------|--------------|----------|
|-------|----------|-----------|-----------|-----------|-----------|-------|-------|-------|-------|--------------|----------|

### Field combinations 3 of 6

[illegible]

[illegible]

[illegible]

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$         | $F B \mu\nu$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | $F H$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | $F W \mu\nu$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | $H e$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      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|-------|----------|-----------|----------------|-----------|---------------|-------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------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------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0             | <div><math display="block">\begin{aligned} &amp; -2\sqrt{\hbar}\log\left(\frac{\hbar\text{bar}^2}{M_F^2}\right)\frac{\overline{\mathcal{P}}_{T_{-1,-7},T_{+2,-9}}H_{e_1}H_{d_1}H_{c_1}B w(0)ps(SU(2))_{L_{e_1,B} w(0)ps(SU(2))_{L_{d_1,c_1}}} }{3M_F} \\ &amp; -2\hbar\log\left(\frac{\hbar\text{bar}^2}{M_F^2}\right)\frac{\overline{\mathcal{P}}_{T_{-1,-7},T_{+2,-9}}^{C_P L_{e_2,Y_e^{-1}}}C_P L_{e_2,Y_e^{-1}}H_{d_1}H_{c_2}B w(0)ps(SU(2))_{L_{e_1,B} w(0)ps(SU(2))_{L_{d_1,c_2}}} }{3M_F} \\ &amp; -2\hbar\log\left(\frac{\hbar\text{bar}^2}{M_F^2}\right)\frac{\overline{\mathcal{P}}_{T_{-1,-7},T_{+2,-9}}^{C_P R_{e_2,Y_e^{-1}}}C_P R_{e_2,Y_e^{-1}}H_{d_1}H_{c_2}B w(0)ps(SU(2))_{L_{e_1,B} w(0)ps(SU(2))_{L_{d_1,c_2}}} }{3M_F} \\ &amp; -\frac{4\hbar\log\left(\frac{\hbar\text{bar}^2}{M_F^2}\right)\overline{\mathcal{P}}_{T_{-1,-7},T_{+2,-9}}H_{e_1}S_{c_1}Y_{c_1}Y_{c_2}H_{c_1}^{\dagger}S_{c_2}^{\dagger}}{\sqrt{8}} \\ &amp; -\frac{1}{2}\sqrt{\hbar}\log\left(\frac{\hbar\text{bar}^2}{M_F^2}\right)\frac{\overline{\mathcal{P}}_{T_{-1,-7},T_{+2,-9}}S_{c_2}Y_{c_1,r_1}Bw(TIMinCG)_{L_{e_1,e_2,i_2}}H_{d_1}^{\dagger}H_{c_2}^{\dagger}S_{c_1}^{\dagger}}{\sqrt{8}} \\ &amp; -\frac{1}{2}\sqrt{\hbar}\log\left(\frac{\hbar\text{bar}^2}{M_F^2}\right)\frac{\overline{\mathcal{P}}_{T_{-1,-7},T_{+2,-9}}S_{c_2}Y_{c_1,r_1}Bw(TIMinCG)_{L_{e_1,e_2,i_2}}H_{d_1}^{\dagger}H_{c_2}^{\dagger}S_{c_1}^{\dagger}}{\sqrt{8}} \\ &amp; -2\sqrt{\hbar}\log\left(\frac{\hbar\text{bar}^2}{M_F^2}\right)\frac{\overline{\mathcal{P}}_{T_{-1,-7},T_{+2,-9}}S_{c_1}^{\dagger}S_{c_2}Y_{c_1,r_1}Y_{c_2}H_{c_1}^{\dagger}S_{c_2}^{\dagger}}{\sqrt{8}} \\ &amp; -\frac{1}{2}\sqrt{\hbar}\log\left(\frac{\hbar\text{bar}^2}{M_F^2}\right)\frac{\overline{\mathcal{P}}_{T_{-1,-7},T_{+2,-9}}H_{d_1}H_{c_2}S_{c_1}Y_{c_1,r_1}Y_{c_2}S_{c_1}^{\dagger}S_{c_2}^{\dagger}}{\sqrt{8}} \\ &amp; -\frac{1}{2}\sqrt{\hbar}\log\left(\frac{\hbar\text{bar}^2}{M_F^2}\right)\frac{\overline{\mathcal{P}}_{T_{-1,-7},T_{+2,-9}}H_{d_1}H_{c_2}S_{c_1}Y_{c_1,r_1}Y_{c_2}S_{c_1}^{\dagger}S_{c_2}^{\dagger}}{\sqrt{8}} \\ &amp; -2\hbar\log\left(\frac{\hbar\text{bar}^2}{M_F^2}\right)\frac{\overline{\mathcal{P}}_{T_{-1,-7},T_{+2,-9}}H_{e_1}S_{c_1}Y_{c_1}Y_{c_2}H_{c_1}^{\dagger}S_{c_2}^{\dagger}}{\sqrt{8}} \\ &amp; -2\hbar\log\left(\frac{\hbar\text{bar}^2}{M_F^2}\right)\frac{\overline{\mathcal{P}}_{T_{-1,-7},T_{+2,-9}}H_{e_2}S_{c_2}Y_{c_2}Y_{c_1}H_{c_2}^{\dagger}S_{c_1}^{\dagger}}{\sqrt{8}} \\ &amp; -\sqrt{\hbar}\log\left(\frac{\hbar\text{bar}^2}{M_F^2}\right)\frac{\overline{\mathcal{P}}_{T_{-1,-7},T_{+2,-9}}H_{d_2}S_{c_2}Y_{c_1,r_1}H_{d_1}^{\dagger}S_{c_1}^{\dagger}Z_{d_1,e_1,e_2,i_2}}{}^{\dagger}Bw^{\dagger}}{\sqrt{8}} \\ &amp; -\sqrt{\hbar}\log\left(\frac{\hbar\text{bar}^2}{M_F^2}\right)\frac{\overline{\mathcal{P}}_{T_{-1,-7},T_{+2,-9}}H_{d_2}S_{c_2}Y_{c_1,r_1}H_{d_1}^{\dagger}S_{c_1}^{\dagger}Z_{d_1,e_1,e_2,i_2}}{}^{\dagger}Bw^{\dagger}}{\sqrt{8}} \\ &amp; -\sqrt{\hbar}\log\left(\frac{\hbar\text{bar}^2}{M_F^2}\right)\frac{\overline{\mathcal{P}}_{T_{-1,-7},T_{+2,-9}}H_{d_2}S_{c_2}Y_{c_1,r_1}H_{d_1}^{\dagger}S_{c_1}^{\dagger}Z_{d_1,e_1,e_2,i_2}}{}^{\dagger}Bw^{\dagger}}{\sqrt{8}} \\ &amp; -\sqrt{\hbar}\log\left(\frac{\hbar\text{bar}^2}{M_F^2}\right)\frac{\overline{\mathcal{P}}_{T_{-1,-7},T_{+2,-9}}H_{d_2}S_{c_2}Y_{c_1,r_1}H_{d_1}^{\dagger}S_{c_1}^{\dagger}Z_{d_1,e_1,e_2,i_2}}{}^{\dagger}Bw^{\dagger}}{\sqrt{8}} \\ &amp; +\sqrt{\hbar}\log\left(\frac{\hbar\text{bar}^2}{M_F^2}\right)\frac{\overline{\mathcal{P}}_{T_{-1,-7},T_{+2,-9}}H_{e_1}S_{c_1}S_{c_2}Y_{c_1,r_1}Bw(0)_{L_{e_1,e_2,i_2}}H_{c_2}^{\dagger}Bw^{\dagger}}{\sqrt{8}} \\ &amp; +\sqrt{\hbar}\log\left(\frac{\hbar\text{bar}^2}{M_F^2}\right)\frac{\overline{\mathcal{P}}_{T_{-1,-7},T_{+2,-9}}H_{e_1}S_{c_1}S_{c_2}Y_{c_1,r_1}Bw(0)_{L_{e_1,e_2,i_2}}H_{c_2}^{\dagger}Bw^{\dagger}}{\sqrt{8}} \\ &amp; +\sqrt{\hbar}\log\left(\frac{\hbar\text{bar}^2}{M_F^2}\right)\frac{\overline{\mathcal{P}}_{T_{-1,-7},T_{+2,-9}}H_{e_2}S_{c_2}S_{c_1}Y_{c_2}H_{c_2}^{\dagger}S_{c_1}^{\dagger}Bw^{\dagger}}{\sqrt{8}} \\ &amp; -\sqrt{\hbar}\log\left(\frac{\hbar\text{bar}^2}{M_F^2}\right)\frac{\overline{\mathcal{P}}_{T_{-1,-7},T_{+2,-9}}H_{e_2}S_{c_2}S_{c_1}Y_{c_2}H_{c_2}^{\dagger}S_{c_1}^{\dagger}Bw^{\dagger}}{\sqrt{8}} \end{aligned}</math></div> | <div><math display="block">-\hbar\log\left(\frac{\mu\text{bar}^2}{M_F^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2</math></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <div><math display="block">\begin{aligned} &amp; +\frac{\sqrt{3}\hbar y_{1,i_3,r_1}\overline{L_{A_1,i_2}}PR e_{i_1}H_{A_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon} \\ &amp; +\frac{3}{8}\frac{\sqrt{3}\hbar y_{1,i_3,r_1}\overline{L_{A_1,i_2}}PR e_{i_1}H_{A_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon} \\ &amp; +\frac{\sqrt{3}\hbar y_{2,i_3,r_1}\overline{L_{A_1,i_2}}PR e_{i_1}H_{A_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon} \\ &amp; +\frac{3}{8}\frac{\sqrt{3}\hbar y_{2,i_3,r_1}\overline{L_{A_1,i_2}}PR e_{i_1}H_{A_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon} \\ &amp; +\frac{\sqrt{3}\hbar y_{1,i_1,r_1}\overline{Y_{e,i_3,i_2}}e_{i_2}PL_{A_1,i_1}y_{1,i_3,r_1}H_{A_1}^{\dagger}}{4\epsilon} \\ &amp; +\frac{3}{8}\frac{\sqrt{3}\hbar y_{1,i_1,r_1}\overline{Y_{e,i_3,i_2}}e_{i_2}PL_{A_1,i_1}y_{1,i_3,r_1}H_{A_1}^{\dagger}}{4\epsilon} \\ &amp; +\frac{\sqrt{3}\hbar y_{2,i_1,r_1}\overline{Y_{e,i_3,i_2}}e_{i_2}PL_{A_1,i_1}y_{2,i_3,r_1}H_{A_1}^{\dagger}}{4\epsilon} \\ &amp; +\frac{3}{8}\frac{\sqrt{3}\hbar y_{2,i_1,r_1}\overline{Y_{e,i_3,i_2}}e_{i_2}PL_{A_1,i_1}y_{2,i_3,r_1}H_{A_1}^{\dagger}}{4\epsilon} \end{aligned}</math></div> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$ | <div><math display="block">-\frac{3}{4}\hbar\log\left(\frac{\mu\text{bar}^2}{M_F^2}\right)B_{\mu_1\mu_2}^2</math></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <div><math display="block">\begin{aligned} &amp; \frac{2\hbar\log\left(\frac{\hbar\text{bar}^2}{M_F^2}\right)\overline{\mathcal{P}}_{T_{-1,-7},T_{+2,-9}}^{C_P L_{e_2,Y_e^{-1}}}C_P L_{e_2,Y_e^{-1}}H_{d_1}H_{c_2}B w(0)ps(SU(2))_{L_{e_1,B} w(0)ps(SU(2))_{L_{d_1,c_2}}} }{3M_F} \\ &amp; -2\hbar\log\left(\frac{\hbar\text{bar}^2}{M_F^2}\right)\frac{\overline{\mathcal{P}}_{T_{-1,-7},T_{+2,-9}}^{C_P R_{e_2,Y_e^{-1}}}C_P R_{e_2,Y_e^{-1}}H_{d_1}H_{c_2}B w(0)ps(SU(2))_{L_{e_1,B} w(0)ps(SU(2))_{L_{d_1,c_2}}} }{3M_F} \\ &amp; -\frac{4\hbar\log\left(\frac{\hbar\text{bar}^2}{M_F^2}\right)\overline{\mathcal{P}}_{T_{-1,-7},T_{+2,-9}}H_{e_1}S_{c_1}Y_{c_1}Y_{c_2}H_{c_1}^{\dagger}S_{c_2}^{\dagger}}{\sqrt{8}} \\ &amp; -\frac{1}{2}\sqrt{\hbar}\log\left(\frac{\hbar\text{bar}^2}{M_F^2}\right)\frac{\overline{\mathcal{P}}_{T_{-1,-7},T_{+2,-9}}S_{c_2}Y_{c_1,r_1}Bw(TIMinCG)_{L_{e_1,e_2,i_2}}H_{d_1}^{\dagger}H_{c_2}^{\dagger}S_{c_1}^{\dagger}}{\sqrt{8}} \\ &amp; -\frac{1}{2}\sqrt{\hbar}\log\left(\frac{\hbar\text{bar}^2}{M_F^2}\right)\frac{\overline{\mathcal{P}}_{T_{-1,-7},T_{+2,-9}}S_{c_2}Y_{c_1,r_1}Bw(TIMinCG)_{L_{e_1,e_2,i_2}}H_{d_1}^{\dagger}H_{c_2}^{\dagger}S_{c_1}^{\dagger}}{\sqrt{8}} \\ &amp; -\frac{1}{2}\sqrt{\hbar}\log\left(\frac{\hbar\text{bar}^2}{M_F^2}\right)\frac{\overline{\mathcal{P}}_{T_{-1,-7},T_{+2,-9}}H_{d_1}H_{c_2}S_{c_1}Y_{c_1,r_1}Y_{c_2}S_{c_1}^{\dagger}S_{c_2}^{\dagger}}{\sqrt{8}} \\ &amp; -\frac{1}{2}\sqrt{\hbar}\log\left(\frac{\hbar\text{bar}^2}{M_F^2}\right)\frac{\overline{\mathcal{P}}_{T_{-1,-7},T_{+2,-9}}H_{d_1}H_{c_2}S_{c_1}Y_{c_1,r_1}Y_{c_2}S_{c_1}^{\dagger}S_{c_2}^{\dagger}}{\sqrt{8}} \\ &amp; -2\hbar\log\left(\frac{\hbar\text{bar}^2}{M_F^2}\right)\frac{\overline{\mathcal{P}}_{T_{-1,-7},T_{+2,-9}}H_{e_1}S_{c_1}Y_{c_1}Y_{c_2}H_{c_1}^{\dagger}S_{c_2}^{\dagger}}{\sqrt{8}} \\ &amp; -2\hbar\log\left(\frac{\hbar\text{bar}^2}{M_F^2}\right)\frac{\overline{\mathcal{P}}_{T_{-1,-7},T_{+2,-9}}H_{e_2}S_{c_2}Y_{c_2}Y_{c_1}H_{c_2}^{\dagger}S_{c_1}^{\dagger}}{\sqrt{8}} \\ &amp; -\sqrt{\hbar}\log\left(\frac{\hbar\text{bar}^2}{M_F^2}\right)\frac{\overline{\mathcal{P}}_{T_{-1,-7},T_{+2,-9}}H_{d_2}S_{c_2}Y_{c_1,r_1}H_{d_1}^{\dagger}S_{c_1}^{\dagger}Z_{d_1,e_1,e_2,i_2}}{}^{\dagger}Bw^{\dagger}}{\sqrt{8}} \\ &amp; -\sqrt{\hbar}\log\left(\frac{\hbar\text{bar}^2}{M_F^2}\right)\frac{\overline{\mathcal{P}}_{T_{-1,-7},T_{+2,-9}}H_{d_2}S_{c_2}Y_{c_1,r_1}H_{d_1}^{\dagger}S_{c_1}^{\dagger}Z_{d_1,e_1,e_2,i_2}}{}^{\dagger}Bw^{\dagger}}{\sqrt{8}} \\ &amp; -\sqrt{\hbar}\log\left(\frac{\hbar\text{bar}^2}{M_F^2}\right)\frac{\overline{\mathcal{P}}_{T_{-1,-7},T_{+2,-9}}H_{d_2}S_{c_2}Y_{c_1,r_1}H_{d_1}^{\dagger}S_{c_1}^{\dagger}Z_{d_1,e_1,e_2,i_2}}{}^{\dagger}Bw^{\dagger}}{\sqrt{8}} \\ &amp; -\sqrt{\hbar}\log\left(\frac{\hbar\text{bar}^2}{M_F^2}\right)\frac{\overline{\mathcal{P}}_{T_{-1,-7},T_{+2,-9}}H_{d_2}S_{c_2}Y_{c_1,r_1}H_{d_1}^{\dagger}S_{c_1}^{\dagger}Z_{d_1,e_1,e_2,i_2}}{}^{\dagger}Bw^{\dagger}}{\sqrt{8}} \\ &amp; +\sqrt{\hbar}\log\left(\frac{\hbar\text{bar}^2}{M_F^2}\right)\frac{\overline{\mathcal{P}}_{T_{-1,-7},T_{+2,-9}}H_{e_1}S_{c_1}S_{c_2}Y_{c_1,r_1}Bw(0)_{L_{e_1,e_2,i_2}}H_{c_2}^{\dagger}Bw^{\dagger}}{\sqrt{8}} \\ &amp; +\sqrt{\hbar}\log\left(\frac{\hbar\text{bar}^2}{M_F^2}\right)\frac{\overline{\mathcal{P}}_{T_{-1,-7},T_{+2,-9}}H_{e_1}S_{c_1}S_{c_2}Y_{c_1,r_1}Bw(0)_{L_{e_1,e_2,i_2}}H_{c_2}^{\dagger}Bw^{\dagger}}{\sqrt{8}} \\ &amp; +\sqrt{\hbar}\log\left(\frac{\hbar\text{bar}^2}{M_F^2}\right)\frac{\overline{\mathcal{P}}_{T_{-1,-7},T_{+2,-9}}H_{e_2}S_{c_2}S_{c_1}Y_{c_2}H_{c_2}^{\dagger}S_{c_1}^{\dagger}Bw^{\dagger}}{\sqrt{8}} \\ &amp; -\sqrt{\hbar}\log\left(\frac{\hbar\text{bar}^2}{M_F^2}\right)\frac{\overline{\mathcal{P}}_{T_{-1,-7},T_{+2,-9}}H_{e_2}S_{c_2}S_{c_1}Y_{c_2}H_{c_2}^{\dagger}S_{c_1}^{\dagger}Bw^{\dagger}}{\sqrt{8}} \end{aligned}</math></div> | <div><math display="block">-2\hbar\log\left(\frac{\mu\text{bar}^2}{M_F^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2</math></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <div><math display="block">\begin{aligned} &amp; +\frac{\sqrt{3}\hbar y_{1,i_3,r_1}\overline{L_{A_1,i_2}}PR e_{i_1}H_{A_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon} \\ &amp; +\frac{3}{8}\frac{\sqrt{3}\hbar y_{1,i_3,r_1}\overline{L_{A_1,i_2}}PR e_{i_1}H_{A_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon} \\ &amp; +\frac{\sqrt{3}\hbar y_{2,i_3,r_1}\overline{L_{A_1,i_2}}PR e_{i_1}H_{A_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon} \\ &amp; +\frac{3}{8}\frac{\sqrt{3}\hbar y_{2,i_3,r_1}\overline{L_{A_1,i_2}}PR e_{i_1}H_{A_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon} \\ &amp; +\frac{\sqrt{3}\hbar y_{1,i_1,r_1}\overline{Y_{e,i_3,i_2}}e_{i_2}PL_{A_1,i_1}y_{1,i_3,r_1}H_{A_1}^{\dagger}}{4\epsilon} \\ &amp; +\frac{3}{8}\frac{\sqrt{3}\hbar y_{1,i_1,r_1}\overline{Y_{e,i_3,i_2}}e_{i_2}PL_{A_1,i_1}y_{1,i_3,r_1}H_{A_1}^{\dagger}}{4\epsilon} \\ &amp; +\frac{\sqrt{3}\hbar y_{2,i_1,r_1}\overline{Y_{e,i_3,i_2}}e_{i_2}PL_{A_1,i_1}y_{2,i_3,r_1}H_{A_1}^{\dagger}}{4\epsilon} \\ &amp; +\frac{3}{8}\frac{\sqrt{3}\hbar y_{2,i_1,r_1}\overline{Y_{e,i_3,i_2}}e_{i_2}PL_{A_1,i_1}y_{2,i_3,r_1}H_{A_1}^{\dagger}}{4\epsilon} \end{aligned}</math></div> |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1             | <div><math display="block">-3\hbar\log\left(\frac{\mu\text{bar}^2}{M_F^2}\right)B_{\mu_1\mu_2}^2</math></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <div><math display="block">\begin{aligned} &amp; \frac{2\hbar\log\left(\frac{\hbar\text{bar}^2}{M_F^2}\right)\overline{\mathcal{P}}_{T_{-1,-7},T_{+2,-9}}^{C_P L_{e_2,Y_e^{-1}}}C_P L_{e_2,Y_e^{-1}}H_{d_1}H_{c_2}B w(0)ps(SU(2))_{L_{e_1,B} w(0)ps(SU(2))_{L_{d_1,c_2}}} }{3M_F} \\ &amp; -2\hbar\log\left(\frac{\hbar\text{bar}^2}{M_F^2}\right)\frac{\overline{\mathcal{P}}_{T_{-1,-7},T_{+2,-9}}^{C_P R_{e_2,Y_e^{-1}}}C_P R_{e_2,Y_e^{-1}}H_{d_1}H_{c_2}B w(0)ps(SU(2))_{L_{e_1,B} w(0)ps(SU(2))_{L_{d_1,c_2}}} }{3M_F} \\ &amp; -\frac{4\hbar\log\left(\frac{\hbar\text{bar}^2}{M_F^2}\right)\overline{\mathcal{P}}_{T_{-1,-7},T_{+2,-9}}H_{e_1}S_{c_1}Y_{c_1}Y_{c_2}H_{c_1}^{\dagger}S_{c_2}^{\dagger}}{\sqrt{8}} \\ &amp; -\frac{1}{2}\sqrt{\hbar}\log\left(\frac{\hbar\text{bar}^2}{M_F^2}\right)\frac{\overline{\mathcal{P}}_{T_{-1,-7},T_{+2,-9}}S_{c_2}Y_{c_1,r_1}Bw(TIMinCG)_{L_{e_1,e_2,i_2}}H_{d_1}^{\dagger}H_{c_2}^{\dagger}S_{c_1}^{\dagger}}{\sqrt{8}} \\ &amp; -\frac{1}{2}\sqrt{\hbar}\log\left(\frac{\hbar\text{bar}^2}{M_F^2}\right)\frac{\overline{\mathcal{P}}_{T_{-1,-7},T_{+2,-9}}S_{c_2}Y_{c_1,r_1}Bw(TIMinCG)_{L_{e_1,e_2,i_2}}H_{d_1}^{\dagger}H_{c_2}^{\dagger}S_{c_1}^{\dagger}}{\sqrt{8}} \\ &amp; -\frac{1}{2}\sqrt{\hbar}\log\left(\frac{\hbar\text{bar}^2}{M_F^2}\right)\frac{\overline{\mathcal{P}}_{T_{-1,-7},T_{+2,-9}}H_{d_1}H_{c_2}S_{c_1}Y_{c_1,r_1}Y_{c_2}S_{c_1}^{\dagger}S_{c_2}^{\dagger}}{\sqrt{8}} \\ &amp; -\frac{1}{2}\sqrt{\hbar}\log\left(\frac{\hbar\text{bar}^2}{M_F^2}\right)\frac{\overline{\mathcal{P}}_{T_{-1,-7},T_{+2,-9}}H_{d_1}H_{c_2}S_{c_1}Y_{c_1,r_1}Y_{c_2}S_{c_1}^{\dagger}S_{c_2}^{\dagger}}{\sqrt{8}} \\ &amp; -2\hbar\log\left(\frac{\hbar\text{bar}^2}{M_F^2}\right)\frac{\overline{\mathcal{P}}_{T_{-1,-7},T_{+2,-9}}H_{e_1}S_{c_1}Y_{c_1}Y_{c_2}H_{c_1}^{\dagger}S_{c_2}^{\dagger}}{\sqrt{8}} \\ &amp; -2\hbar\log\left(\frac{\hbar\text{bar}^2}{M_F^2}\right)\frac{\overline{\mathcal{P}}_{T_{-1,-7},T_{+2,-9}}H_{e_2}S_{c_2}Y_{c_2}Y_{c_1}H_{c_2}^{\dagger}S_{c_1}^{\dagger}}{\sqrt{8}} \\ &amp; -\sqrt{\hbar}\log\left(\frac{\hbar\text{bar}^2}{M_F^2}\right)\frac{\overline{\mathcal{P}}_{T_{-1,-7},T_{+2,-9}}H_{d_2}S_{c_2}Y_{c_1,r_1}H_{d_1}^{\dagger}S_{c_1}^{\dagger}Z_{d_1,e_1,e_2,i_2}}{}^{\dagger}Bw^{\dagger}}{\sqrt{8}} \\ &amp; -\sqrt{\hbar}\log\left(\frac{\hbar\text{bar}^2}{M_F^2}\right)\frac{\overline{\mathcal{P}}_{T_{-1,-7},T_{+2,-9}}H_{d_2}S_{c_2}Y_{c_1,r_1}H_{d_1}^{\dagger}S_{c_1}^{\dagger}Z_{d_1,e_1,e_2,i_2}}{}^{\dagger}Bw^{\dagger}}{\sqrt{8}} \\ &amp; -\sqrt{\hbar}\log\left(\frac{\hbar\text{bar}^2}{M_F^2}\right)\frac{\overline{\mathcal{P}}_{T_{-1,-7},T_{+2,-9}}H_{d_2}S_{c_2}Y_{c_1,r_1}H_{d_1}^{\dagger}S_{c_1}^{\dagger}Z_{d_1,e_1,e_2,i_2}}{}^{\dagger}Bw^{\dagger}}{\sqrt{8}} \\ &amp; -\sqrt{\hbar}\log\left(\frac{\hbar\text{bar}^2}{M_F^2}\right)\frac{\overline{\mathcal{P}}_{T_{-1,-7},T_{+2,-9}}H_{d_2}S_{c_2}Y_{c_1,r_1}H_{d_1}^{\dagger}S_{c_1}^{\dagger}Z_{d_1,e_1,e_2,i_2}}{}^{\dagger}Bw^{\dagger}}{\sqrt{8}} \\ &amp; +\sqrt{\hbar}\log\left(\frac{\hbar\text{bar}^2}{M_F^2}\right)\frac{\overline{\mathcal{P}}_{T_{-1,-7},T_{+2,-9}}H_{e_1}S_{c_1}S_{c_2}Y_{c_1,r_1}Bw(0)_{L_{e_1,e_2,i_2}}H_{c_2}^{\dagger}Bw^{\dagger}}{\sqrt{8}} \\ &amp; +\sqrt{\hbar}\log\left(\frac{\hbar\text{bar}^2}{M_F^2}\right)\frac{\overline{\mathcal{P}}_{T_{-1,-7},T_{+2,-9}}H_{e_1}S_{c_1}S_{c_2}Y_{c_1,r_1}Bw(0)_{L_{e_1,e_2,i_2}}H_{c_2}^{\dagger}Bw^{\dagger}}{\sqrt{8}} \\ &amp; +\sqrt{\hbar}\log\left(\frac{\hbar\text{bar}^2}{M_F^2}\right)\frac{\overline{\mathcal{P}}_{T_{-1,-7},T_{+2,-9}}H_{e_2}S_{c_2}S_{c_1}Y_{c_2}H_{c_2}^{\dagger}S_{c_1}^{\dagger}Bw^{\dagger}}{\sqrt{8}} \\ &amp; -\sqrt{\hbar}\log\left(\frac{\hbar\text{bar}^2}{M_F^2}\right)\frac{\overline{\mathcal{P}}_{T_{-1,-7},T_{+2,-9}}H_{e_2}S_{c_2}S_{c_1}Y_{c_2}H_{c_2}^{\dagger}S_{c_1}^{\dagger}Bw^{\dagger}}{\sqrt{8}} \end{aligned}</math></div> | <div><math display="block">-2\hbar\log\left(\frac{\mu\text{bar}^2}{M_F^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2</math></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <div><math display="block">\begin{aligned} &amp; +\frac{\sqrt{3}\hbar y_{1,i_3,r_1}\overline{L_{A_1,i_2}}PR e_{i_1}H_{A_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon} \\ &amp; +\frac{3}{8}\frac{\sqrt{3}\hbar y_{1,i_3,r_1}\overline{L_{A_1,i_2}}PR e_{i_1}H_{A_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon} \\ &amp; +\frac{\sqrt{3}\hbar y_{2,i_3,r_1}\overline{L_{A_1,i_2}}PR e_{i_1}H_{A_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon} \\ &amp; +\frac{3}{8}\frac{\sqrt{3}\hbar y_{2,i_3,r_1}\overline{L_{A_1,i_2}}PR e_{i_1}H_{A_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon} \\ &amp; +\frac{\sqrt{3}\hbar y_{1,i_1,r_1}\overline{Y_{e,i_3,i_2}}e_{i_2}PL_{A_1,i_1}y_{1,i_3,r_1}H_{A_1}^{\dagger}}{4\epsilon} \\ &amp; +\frac{3}{8}\frac{\sqrt{3}\hbar y_{1,i_1,r_1}\overline{Y_{e,i_3,i_2}}e_{i_2}PL_{A_1,i_1}y_{1,i_3,r_1}H_{A_1}^{\dagger}}{4\epsilon} \\ &amp; +\frac{\sqrt{3}\hbar y_{2,i_1,r_1}\overline{Y_{e,i_3,i_2}}e_{i_2}PL_{A_1,i_1}y_{2,i_3,r_1}H_{A_1}^{\dagger}}{4\epsilon} \\ &amp; +\frac{3}{8}\frac{\sqrt{3}\hbar y_{2,i_1,r_1}\overline{Y_{e,i_3,i_2}}e_{i_2}PL_{A_1,i_1}y_{2,i_3,r_1}H_{A_1}^{\dagger}}{4\epsilon} \end{aligned}</math></div> |

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## Field combinations 4 of 6

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## Field combinations 5 of 6

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | $S_2 F$ | $S_2 H$ | $F H e$ | $S_1 F H$                                                                                                                                                                                       |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------|---------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ |         |         |         | $-2h \log \left( \frac{\mu \text{bar}^2}{M_F^2} \right) \frac{1}{\overline{y}_{1,1,\tau_1} \overline{y}_{e,2,12} H_{a_1} S_1 y_{1,2,\tau_1} Y_{e,1,13} H_{a_1}^{-1} S_1^{-1}}$                  |
|       |          |           |                |           |               |       |                |         |         |         | $+\frac{1}{4} h \log \left( \frac{\mu \text{bar}^2}{M_F^2} \right) \frac{1}{\overline{y}_{1,1,3,\tau_1} \overline{L}_{a_1,12} P_R e_{11} H_{a_1} y_{1,12,\tau_1} Y_{e,1,3,11}}$                 |
|       |          |           |                |           |               |       |                |         |         |         | $+\frac{1}{4} \sqrt{3} h \log \left( \frac{\mu \text{bar}^2}{M_F^2} \right) \frac{1}{\overline{y}_{2,1,3,\tau_1} \overline{L}_{a_1,12} P_R e_{11} H_{a_1} y_{2,12,\tau_1} Y_{e,1,3,11}}$        |
|       |          |           |                |           |               |       |                |         |         |         | $+\frac{1}{4} h \log \left( \frac{\mu \text{bar}^2}{M_F^2} \right) \frac{1}{\overline{y}_{1,1,1,\tau_1} \overline{Y}_{e,1,3,12} \overline{e}_{12} P_L L_{a_1,11} y_{1,13,\tau_1} H_{a_1}^{-1}}$ |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              |         |         |         | $-h \log \left( \frac{\mu \text{bar}^2}{M_F^2} \right) \frac{1}{\overline{y}_{1,1,\tau_1} \overline{Y}_{e,2,12} H_{a_1} S_1 y_{1,2,\tau_1} Y_{e,1,13} H_{a_1}^{-1} S_1^{-1}}$                   |
|       |          |           |                |           |               |       |                |         |         |         | $+\frac{1}{4} h \log \left( \frac{\mu \text{bar}^2}{M_F^2} \right) \frac{1}{\overline{y}_{1,1,3,\tau_1} \overline{L}_{a_1,12} P_R e_{11} H_{a_1} y_{1,12,\tau_1} Y_{e,1,3,11}}$                 |
|       |          |           |                |           |               |       |                |         |         |         | $+\frac{1}{4} \sqrt{3} h \log \left( \frac{\mu \text{bar}^2}{M_F^2} \right) \frac{1}{\overline{y}_{2,1,3,\tau_1} \overline{L}_{a_1,12} P_R e_{11} H_{a_1} y_{2,12,\tau_1} Y_{e,1,3,11}}$        |
|       |          |           |                |           |               |       |                |         |         |         | $+\frac{1}{4} h \log \left( \frac{\mu \text{bar}^2}{M_F^2} \right) \frac{1}{\overline{y}_{1,1,1,\tau_1} \overline{Y}_{e,1,3,12} \overline{e}_{12} P_L L_{a_1,11} y_{1,13,\tau_1} H_{a_1}^{-1}}$ |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  |         |         |         | $-2h \log \left( \frac{\mu \text{bar}^2}{M_F^2} \right) \frac{1}{\overline{y}_{1,1,\tau_1} \overline{Y}_{e,2,12} H_{a_1} S_1 y_{1,2,\tau_1} Y_{e,1,13} H_{a_1}^{-1} S_1^{-1}}$                  |
|       |          |           |                |           |               |       |                |         |         |         | $+\frac{1}{4} h \log \left( \frac{\mu \text{bar}^2}{M_F^2} \right) \frac{1}{\overline{y}_{1,1,3,\tau_1} \overline{L}_{a_1,12} P_R e_{11} H_{a_1} y_{1,12,\tau_1} Y_{e,1,3,11}}$                 |
|       |          |           |                |           |               |       |                |         |         |         | $+\frac{1}{4} \sqrt{3} h \log \left( \frac{\mu \text{bar}^2}{M_F^2} \right) \frac{1}{\overline{y}_{2,1,3,\tau_1} \overline{L}_{a_1,12} P_R e_{11} H_{a_1} y_{2,12,\tau_1} Y_{e,1,3,11}}$        |
|       |          |           |                |           |               |       |                |         |         |         | $+\frac{1}{4} h \log \left( \frac{\mu \text{bar}^2}{M_F^2} \right) \frac{1}{\overline{y}_{1,1,1,\tau_1} \overline{Y}_{e,1,3,12} \overline{e}_{12} P_L L_{a_1,11} y_{1,13,\tau_1} H_{a_1}^{-1}}$ |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              |         |         |         | $-2h \log \left( \frac{\mu \text{bar}^2}{M_F^2} \right) \frac{1}{\overline{y}_{1,1,\tau_1} \overline{Y}_{e,2,12} H_{a_1} S_1 y_{1,2,\tau_1} Y_{e,1,13} H_{a_1}^{-1} S_1^{-1}}$                  |
|       |          |           |                |           |               |       |                |         |         |         | $+\frac{1}{4} h \log \left( \frac{\mu \text{bar}^2}{M_F^2} \right) \frac{1}{\overline{y}_{1,1,3,\tau_1} \overline{L}_{a_1,12} P_R e_{11} H_{a_1} y_{1,12,\tau_1} Y_{e,1,3,11}}$                 |
|       |          |           |                |           |               |       |                |         |         |         | $+\frac{1}{4} \sqrt{3} h \log \left( \frac{\mu \text{bar}^2}{M_F^2} \right) \frac{1}{\overline{y}_{2,1,3,\tau_1} \overline{L}_{a_1,12} P_R e_{11} H_{a_1} y_{2,12,\tau_1} Y_{e,1,3,11}}$        |
|       |          |           |                |           |               |       |                |         |         |         | $+\frac{1}{4} h \log \left( \frac{\mu \text{bar}^2}{M_F^2} \right) \frac{1}{\overline{y}_{1,1,1,\tau_1} \overline{Y}_{e,1,3,12} \overline{e}_{12} P_L L_{a_1,11} y_{1,13,\tau_1} H_{a_1}^{-1}}$ |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  |         |         |         | $-2h \log \left( \frac{\mu \text{bar}^2}{M_F^2} \right) \frac{1}{\overline{y}_{1,1,\tau_1} \overline{Y}_{e,2,12} H_{a_1} S_1 y_{1,2,\tau_1} Y_{e,1,13} H_{a_1}^{-1} S$                          |

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| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | $S_2$ | $F$ | $S_2$ | $H$ | $F$ | $H$ | $e$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | $S_1$ | $F$ | $H$ |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-------|-----|-------|-----|-----|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----|-----|
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  |       |     |       |     |     |     | $+\frac{1}{4}\sqrt{3}h\log\left(\frac{\mu\text{bar}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_2}}P_Re_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{1}{4}\sqrt{3}h\log\left(\frac{\mu\text{bar}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_2}}P_Re_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{1}{4}\sqrt{3}h\log\left(\frac{\mu\text{bar}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}\overline{e_{i_2}}}P_LL_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$ $+\frac{1}{4}\sqrt{3}h\log\left(\frac{\mu\text{bar}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}\overline{e_{i_2}}}P_LL_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$ |       |     |     |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ |       |     |       |     |     |     | $+\frac{1}{4}\sqrt{3}h\log\left(\frac{\mu\text{bar}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_2}}P_Re_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{1}{4}h\log\left(\frac{\mu\text{bar}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_2}}P_Re_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{1}{4}\sqrt{3}h\log\left(\frac{\mu\text{bar}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}\overline{e_{i_2}}}P_LL_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$ $+\frac{1}{4}h\log\left(\frac{\mu\text{bar}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}\overline{e_{i_2}}}P_LL_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$                 |       |     |     |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              |       |     |       |     |     |     | $+\frac{1}{4}\sqrt{3}h\log\left(\frac{\mu\text{bar}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_2}}P_Re_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{1}{4}h\log\left(\frac{\mu\text{bar}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_2}}P_Re_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{1}{4}\sqrt{3}h\log\left(\frac{\mu\text{bar}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}\overline{e_{i_2}}}P_LL_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$ $+\frac{1}{4}h\log\left(\frac{\mu\text{bar}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}\overline{e_{i_2}}}P_LL_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$                 |       |     |     |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$     | $Y_{S_2}$ | $d_F$         | $Y_F$         | $S_2 F$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | $S_2 H$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | $F H e$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | $S_1 F H$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------|----------|-----------|-----------|---------------|-----------|---------------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------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| T3-D  | 0        | 3         | 0         | 1             | 1         | 2             | $\frac{1}{2}$ | $\begin{aligned} & -2\lambda \log \left( \frac{\mu \text{bar}^2}{M_F^2} \right) \overline{W}_{r_1,r_1} W_{r_2,r_2} S_{r_1,r_2}^2 W_{r_1,r_2} \left( S_{r_1'}^\dagger \right)^2 \\ & -2\sqrt{N_c} w_8 i s t r 22 h \log \left( \frac{\mu \text{bar}^2}{M_F^2} \right) \overline{W}_{r_1,r_1} S_{r_1} S_{r_2} W_{r_1,r_2} \\ & \frac{\sqrt{3} h \log \left( \frac{\mu \text{bar}^2}{M_F^2} \right) \overline{W}_{r_1,r_1} L_{r_1,r_1} C P_R \overline{L}_{r_2,r_2}^\dagger S_{r_1,r_1} W_{r_1,r_2} W_{r_1,r_2} B ar(T3MixCG)_{r_1,r_2,j_1} S_{r_1}^\dagger}{4 M_F} \\ & h \log \left( \frac{\mu \text{bar}^2}{M_F^2} \right) \overline{W}_{r_1,r_1} L_{r_1,r_1} C P_R \overline{L}_{r_2,r_2}^\dagger S_{r_1,r_1} W_{r_1,r_2} W_{r_1,r_2} B ar(T3MixCG)_{r_1,r_2,j_1} S_{r_1}^\dagger \\ & h \log \left( \frac{\mu \text{bar}^2}{M_F^2} \right) \overline{W}_{r_1,r_1} L_{r_1,r_1} C P_R \overline{L}_{r_2,r_2}^\dagger S_{r_1,r_1} W_{r_1,r_2} W_{r_1,r_2} B ar(T3MixCG)_{r_1,r_2,j_1} S_{r_1}^\dagger \\ & h \log \left( \frac{\mu \text{bar}^2}{M_F^2} \right) \overline{W}_{r_1,r_1} L_{r_1,r_1} C P_R \overline{L}_{r_2,r_2}^\dagger S_{r_1,r_1} W_{r_1,r_2} W_{r_1,r_2} B ar(T3MixCG)_{r_1,r_2,j_1} S_{r_1}^\dagger \\ & \frac{\sqrt{3} h \log \left( \frac{\mu \text{bar}^2}{M_F^2} \right) \overline{W}_{r_1,r_1} L_{r_1,r_1} C P_R \overline{L}_{r_2,r_2}^\dagger S_{r_1,r_1} W_{r_1,r_2} W_{r_1,r_2} B ar(T3MixCG)_{r_1,r_2,j_1} S_{r_1}^\dagger}{4 M_F} \\ & h \log \left( \frac{\mu \text{bar}^2}{M_F^2} \right) \overline{W}_{r_1,r_1} L_{r_1,r_1} C P_R \overline{L}_{r_2,r_2}^\dagger S_{r_1,r_1} W_{r_1,r_2} W_{r_1,r_2} B ar(T3MixCG)_{r_1,r_2,j_1} S_{r_1}^\dagger \\ & \frac{\sqrt{3} h \log \left( \frac{\mu \text{bar}^2}{M_F^2} \right) \overline{W}_{r_1,r_1} W_{r_1,r_2} W_{r_1,r_2} L_{r_2,r_2}^\dagger C P_R L_{r_1,r_1} S_{2H,r_1,r_2} T_{r_1,r_2} S_{r_1,r_2}^\dagger S_{r_1,r_2}^\dagger}{4 M_F} \\ & h \log \left( \frac{\mu \text{bar}^2}{M_F^2} \right) \overline{W}_{r_1,r_1} W_{r_1,r_2} W_{r_1,r_2} L_{r_2,r_2}^\dagger C P_R L_{r_1,r_1} S_{2H,r_1,r_2} T_{r_1,r_2} S_{r_1,r_2}^\dagger S_{r_1,r_2}^\dagger \\ & h \log \left( \frac{\mu \text{bar}^2}{M_F^2} \right) \overline{W}_{r_1,r_1} W_{r_1,r_2} W_{r_1,r_2} L_{r_2,r_2}^\dagger C P_R L_{r_1,r_1} S_{2H,r_1,r_2} T_{r_1,r_2} S_{r_1,r_2}^\dagger S_{r_1,r_2}^\dagger \\ & \frac{4 h \log \left( \frac{\mu \text{bar}^2}{M_F^2} \right) \overline{W}_{r_1,r_1} W_{r_1,r_2} W_{r_1,r_2} S_{r_1,r_2}^\dagger S_{r_1,r_2}^\dagger S_{r_1,r_2}^\dagger S_{r_1,r_2}^\dagger}{\sqrt{3}} \\ & h \log \left( \frac{\mu \text{bar}^2}{M_F^2} \right) L_{r_1,r_1} C P_R \overline{L}_{r_2,r_2}^\dagger S_{r_1,r_1} W_{r_1,r_2} B ar(T3MixCG)_{r_1,r_2,j_1} S_{r_1}^\dagger S_{r_1}^\dagger \\ & \frac{h \log \left( \frac{\mu \text{bar}^2}{M_F^2} \right) \overline{W}_{r_1,r_1} W_{r_1,r_2} W_{r_1,r_2} L_{r_2,r_2}^\dagger C P_R L_{r_1,r_1} S_{2H,r_1,r_2} T_{r_1,r_2} S_{r_1,r_2}^\dagger S_{r_1,r_2}^\dagger}{M_F} \\ & \frac{2 h \log \left( \frac{\mu \text{bar}^2}{M_F^2} \right) \overline{W}_{r_1,r_1} S_{r_1} W_{r_1,r_2} W_{r_1,r_2} S_{r_1}^\dagger S_{r_1}^\dagger T_{r_1,r_2} S_{r_1,r_2}^\dagger S_{r_1,r_2}^\dagger}{\sqrt{3}} \\ & -2 h \log \left( \frac{\mu \text{bar}^2}{M_F^2} \right) \overline{W}_{r_1,r_1} S_{r_1} S_{r_2} W_{r_1,r_1} S_{r_1}^\dagger S_{r_1}^\dagger T_{r_1,r_2} S_{r_1,r_2}^\dagger S_{r_1,r_2}^\dagger \\ & -2 h \log \left( \frac{\mu \text{bar}^2}{M_F^2} \right) \overline{W}_{r_1,r_1} S_{r_1}^2 W_{r_1,r_1} \left( S_{r_1'}^\dagger \right)^2 S_{r_1,r_1}^\dagger \end{aligned}$ | $\begin{aligned} & + H_{a_1} H_{a_2} S_{1,j_1} \overline{T}_{T3,a_1,a_2,j_1} \lambda_{T3} S_2^\dagger \\ & - \frac{h \overline{y}_{1,i_1,r_1} H_{a_1} H_{a_2} S_{1,j_1} y_{1,i_1,r_1} \overline{T}_{T3,a_1,a_2,j_1} \lambda_{T3} S_2^\dagger}{\sqrt{3} \epsilon} \\ & - \frac{h \overline{y}_{1,i_1,r_1} H_{a_1} H_{a_2} S_{1,j_1} y_{1,i_1,r_1} \overline{T}_{T3,a_1,a_2,j_1} \lambda_{T3} S_2^\dagger}{2 \sqrt{3}} \\ & - \frac{h \overline{y}_{2,i_1,r_1} H_{a_1} H_{a_2} S_{1,j_1} y_{2,i_1,r_1} \overline{T}_{T3,a_1,a_2,j_1} \lambda_{T3} S_2^\dagger}{\epsilon} \\ & - \frac{1}{2} h \overline{y}_{2,i_1,r_1} H_{a_1} H_{a_2} S_{1,j_1} y_{2,i_1,r_1} \overline{T}_{T3,a_1,a_2,j_1} \lambda_{T3} S_2^\dagger \\ & - \frac{2 h \overline{y}_{2,i_1,r_1} Y_{e,i_2,i_3} H_{a_1} S_{2Y2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_2^\dagger}{\epsilon} \\ & + \overline{\lambda}_{T3} S_2 \text{Bar}(T3MixCG)_{a_1,a_2,j_1} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger \\ & - \frac{h \overline{y}_{1,i_1,r_1} \overline{\lambda}_{T3} S_2 y_{1,i_1,r_1} \text{Bar}(T3MixCG)_{a_1,a_2,j_1} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\sqrt{3} \epsilon} \\ & - \frac{h \overline{y}_{1,i_1,r_1} \overline{\lambda}_{T3} S_2 y_{1,i_1,r_1} \text{Bar}(T3MixCG)_{a_1,a_2,j_1} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{2 \sqrt{3}} \\ & - \frac{h \overline{y}_{2,i_1,r_1} \overline{\lambda}_{T3} S_2 y_{2,i_1,r_1} \text{Bar}(T3MixCG)_{a_1,a_2,j_1} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\epsilon} \\ & - \frac{1}{2} h \overline{y}_{2,i_1,r_1} \overline{\lambda}_{T3} S_2 y_{2,i_1,r_1} \text{Bar}(T3MixCG)_{a_1,a_2,j_1} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger \\ & + H_{a_2} S_2 H_{a_1}^\dagger S_2^\dagger \overline{\chi}_{H2,a_1,a_2}^\dagger \lambda_{H2}^\dagger \\ & - \frac{2 h \overline{y}_{2,i_1,r_1} H_{a_2} S_2 y_{2,i_1,r_1} H_{a_1}^\dagger S_2^\dagger \overline{\chi}_{H2,a_1,a_2}^\dagger \lambda_{H2}^\dagger}{\epsilon} \\ & - h \overline{y}_{2,i_1,r_1} H_{a_2} S_2 y_{2,i_1,r_1} H_{a_1}^\dagger S_2^\dagger \overline{\chi}_{H2,a_1,a_2}^\dagger \lambda_{H2}^\dagger \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | $\begin{aligned} & + \frac{1}{4} \sqrt{3} h \log \left( \frac{\mu \text{bar}^2}{M_F^2} \right) \overline{y}_{1,i_3,r_1} \overline{L}_{a_1,i_2} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1} \\ & + \frac{1}{4} h \log \left( \frac{\mu \text{bar}^2}{M_F^2} \right) \overline{y}_{2,i_3,r_1} \overline{L}_{a_1,i_2} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1} \\ & + \frac{1}{4} \sqrt{3} h \log \left( \frac{\mu \text{bar}^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} Y_{e,i_3,i_2} \overline{e}_{i_2} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger \\ & + \frac{1}{4} h \log \left( \frac{\mu \text{bar}^2}{M_F^2} \right) \overline{y}_{2,i_1,r_1} Y_{e,i_3,i_2} \overline{e}_{i_2} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|       | T3-D     | 1         | 3         | $\frac{1}{2}$ | 1         | $\frac{3}{2}$ | 2             | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | $\begin{aligned} & -2\lambda \log \left( \frac{\mu \text{bar}^2}{M_F^2} \right) \overline{W}_{r_1,r_1} W_{r_2,r_2} S_{r_1,r_2}^2 W_{r_1,r_2} \left( S_{r_1'}^\dagger \right)^2 \\ & -2\sqrt{N_c} w_8 i s t r 22 h \log \left( \frac{\mu \text{bar}^2}{M_F^2} \right) \overline{W}_{r_1,r_1} S_{r_1} S_{r_2} W_{r_1,r_2} \\ & \frac{\sqrt{3} h \log \left( \frac{\mu \text{bar}^2}{M_F^2} \right) \overline{W}_{r_1,r_1} L_{r_1,r_1} C P_R \overline{L}_{r_2,r_2}^\dagger S_{r_1,r_1} W_{r_1,r_2} W_{r_1,r_2} B ar(T3MixCG)_{r_1,r_2,j_1} S_{r_1}^\dagger}{4 M_F} \\ & 9 h \log \left( \frac{\mu \text{bar}^2}{M_F^2} \right) \overline{W}_{r_1,r_1} L_{r_1,r_1} C P_R \overline{L}_{r_2,r_2}^\dagger S_{r_1,r_1} W_{r_1,r_2} W_{r_1,r_2} B ar(T3MixCG)_{r_1,r_2,j_1} S_{r_1}^\dagger \\ & h \log \left( \frac{\mu \text{bar}^2}{M_F^2} \right) \overline{W}_{r_1,r_1} L_{r_1,r_1} C P_R \overline{L}_{r_2,r_2}^\dagger S_{r_1,r_1} W_{r_1,r_2} W_{r_1,r_2} B ar(T3MixCG)_{r_1,r_2,j_1} S_{r_1}^\dagger \\ & h \log \left( \frac{\mu \text{bar}^2}{M_F^2} \right) \overline{W}_{r_1,r_1} L_{r_1,r_1} C P_R \overline{L}_{r_2,r_2}^\dagger S_{r_1,r_1} W_{r_1,r_2} W_{r_1,r_2} B ar(T3MixCG)_{r_1,r_2,j_1} S_{r_1}^\dagger \\ & h \log \left( \frac{\mu \text{bar}^2}{M_F^2} \right) \overline{W}_{r_1,r_1} L_{r_1,r_1} C P_R \overline{L}_{r_2,r_2}^\dagger S_{r_1,r_1} W_{r_1,r_2} W_{r_1,r_2} B ar(T3MixCG)_{r_1,r_2,j_1} S_{r_1}^\dagger \\ & \frac{\sqrt{3} h \log \left( \frac{\mu \text{bar}^2}{M_F^2} \right) \overline{W}_{r_1,r_1} L_{r_1,r_1} C P_R \overline{L}_{r_2,r_2}^\dagger S_{r_1,r_1} W_{r_1,r_2} W_{r_1,r_2} B ar(T3MixCG)_{r_1,r_2,j_1} S_{r_1}^\dagger}{4 M_F} \\ & h \log \left( \frac{\mu \text{bar}^2}{M_F^2} \right) \overline{W}_{r_1,r_1} L_{r_1,r_1} C P_R \overline{L}_{r_2,r_2}^\dagger S_{r_1,r_1} W_{r_1,r_2} W_{r_1,r_2} B ar(T3MixCG)_{r_1,r_2,j_1} S_{r_1}^\dagger \\ & \frac{\sqrt{3} h \log \left( \frac{\mu \text{bar}^2}{M_F^2} \right) \overline{W}_{r_1,r_1} L_{r_1,r_1} C P_R \overline{L}_{r_2,r_2}^\dagger S_{r_1,r_1} W_{r_1,r_2} W_{r_1,r_2} B ar(T3MixCG)_{r_1,r_2,j_1} S_{r_1}^\dagger}{4 M_F} \\ & h \log \left( \frac{\mu \text{bar}^2}{M_F^2} \right) \overline{W}_{r_1,r_1} L_{r_1,r_1} C P_R \overline{L}_{r_2,r_2}^\dagger S_{r_1,r_1} W_{r_1,r_2} W_{r_1,r_2} B ar(T3MixCG)_{r_1,r_2,j_1} S_{r_1}^\dagger \\ & \frac{\sqrt{3} h \log \left( \frac{\mu \text{bar}^2}{M_F^2} \right) \overline{W}_{r_1,r_1} W_{r_1,r_2} W_{r_1,r_2} L_{r_2,r_2}^\dagger C P_R L_{r_1,r_1} S_{2H,r_1,r_2} T_{r_1,r_2} S_{r_1,r_2}^\dagger S_{r_1,r_2}^\dagger}{4 M_F} \\ & h \log \left( \frac{\mu \text{bar}^2}{M_F^2} \right) \overline{W}_{r_1,r_1} W_{r_1,r_2} W_{r_1,r_2} L_{r_2,r_2}^\dagger C P_R L_{r_1,r_1} S_{2H,r_1,r_2} T_{r_1,r_2} S_{r_1,r_2}^\dagger S_{r_1,r_2}^\dagger \\ & h \log \left( \frac{\mu \text{bar}^2}{M_F^2} \right) \overline{W}_{r_1,r_1} W_{r_1,r_2} W_{r_1,r_2} L_{r_2,r_2}^\dagger C P_R L_{r_1,r_1} S_{2H,r_1,r_2} T_{r_1,r_2} S_{r_1,r_2}^\dagger S_{r_1,r_2}^\dagger \\ & \frac{4 h \log \left( \frac{\mu \text{bar}^2}{M_F^2} \right) \overline{W}_{r_1,r_1} W_{r_1,r_2} W_{r_1,r_2} S_{r_1,r_2}^\dagger S_{r_1,r_2}^\dagger S_{r_1,r_2}^\dagger S_{r_1,r_2}^\dagger}{\sqrt{3}} \\ & h \log \left( \frac{\mu \text{bar}^2}{M_F^2} \right) L_{r_1,r_1} C P_R \overline{L}_{r_2,r_2}^\dagger S_{r_1,r_1} W_{r_1,r_2} B ar(T3MixCG)_{r_1,r_2,j_1} S_{r_1}^\dagger S_{r_1}^\dagger \\ & \frac{h \log \left( \frac{\mu \text{bar}^2}{M_F^2} \right) \overline{W}_{r_1,r_1} W_{r_1,r_2} W_{r_1,r_2} L_{r_2,r_2}^\dagger C P_R L_{r_1,r_1} S_{2H,r_1,r_2} T_{r_1,r_2} S_{r_1,r_2}^\dagger S_{r_1,r_2}^\dagger}{M_F} \\ & \frac{2 h \log \left( \frac{\mu \text{bar}^2}{M_F^2} \right) \overline{W}_{r_1,r_1} S_{r_1} W_{r_1,r_2} W_{r_1,r_2} S_{r_1}^\dagger S_{r_1}^\dagger T_{r_1,r_2} S_{r_1,r_2}^\dagger S_{r_1,r_2}^\dagger}{\sqrt{3}} \\ & -2 h \log \left( \frac{\mu \text{bar}^2}{M_F^2} \right) \overline{W}_{r_1,r_1} S_{r_1} S_{r_2} W_{r_1,r_1} S_{r_1}^\dagger S_{r_1}^\dagger T_{r_1,r_2} S_{r_1,r_2}^\dagger S_{r_1,r_2}^\dagger \\ & -2 h \log \left( \frac{\mu \text{bar}^2}{M_F^2} \right) \overline{W}_{r_1,r_1} S_{r_1}^2 W_{r_1,r_1} \left( S_{r_1'}^\dagger \right)^2 S_{r_1,r_1}^\dagger \end{aligned}$ | $\begin{aligned} & + H_{a_1} H_{a_2} S_{1,j_1} \overline{T}_{T3,a_1,a_2,j_1} \lambda_{T3} S_2^\dagger \\ & - \frac{h \overline{y}_{1,i_1,r_1} H_{a_1} H_{a_2} S_{1,j_1} y_{1,i_1,r_1} \overline{T}_{T3,a_1,a_2,j_1} \lambda_{T3} S_2^\dagger}{\sqrt{3} \epsilon} \\ & - \frac{h \overline{y}_{1,i_1,r_1} H_{a_1} H_{a_2} S_{1,j_1} y_{1,i_1,r_1} \overline{T}_{T3,a_1,a_2,j_1} \lambda_{T3} S_2^\dagger}{2 \sqrt{3}} \\ & - \frac{h \overline{y}_{2,i_1,r_1} H_{a_1} H_{a_2} S_{1,j_1} y_{2,i_1,r_1} \overline{T}_{T3,a_1,a_2,j_1} \lambda_{T3} S_2^\dagger}{\epsilon} \\ & - \frac{1}{2} h \overline{y}_{2,i_1,r_1} H_{a_1} H_{a_2} S_{1,j_1} y_{2,i_1,r_1} \overline{T}_{T3,a_1,a_2,j_1} \lambda_{T3} S_2^\dagger \\ & - \frac{2 h \overline{y}_{2,i_1,r_1} Y_{e,i_2,i_3} H_{a_1} S_{2Y2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_2^\dagger}{\epsilon} \\ & + \overline{\lambda}_{T3} S_2 \text{Bar}(T3MixCG)_{a_1,a_2,j_1} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger \\ & - \frac{h \overline{y}_{1,i_1,r_1} \overline{\lambda}_{T3} S_2 y_{1,i_1,r_1} \text{Bar}(T3MixCG)_{a_1,a_2,j_1} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\sqrt{3} \epsilon} \\ & - \frac{h \overline{y}_{1,i_1,r_1} \overline{\lambda}_{T3} S_2 y_{1,i_1,r_1} \text{Bar}(T3MixCG)_{a_1,a_2,j_1} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{2 \sqrt{3}} \\ & - \frac{h \overline{y}_{2,i_1,r_1} \overline{\lambda}_{T3} S_2 y_{2,i_1,r_1} \text{Bar}(T3MixCG)_{a_1,a_2,j_1} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\epsilon} \\ & - \frac{1}{2} h \overline{y}_{2,i_1,r_1} \overline{\lambda}_{T3} S_2 y_{2,i_1,r_1} \text{Bar}(T3MixCG)_{a_1,a_2,j_1} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger \\ & + H_{a_2} S_2 H_{a_1}^\dagger S_2^\dagger \overline{\chi}_{H2,a_1,a_2}^\dagger \lambda_{H2}^\dagger \\ & - \frac{2 h \overline{y}_{2,i_1,r_1} H_{a_2} S_2 y_{2,i_1,r_1} H_{a_1}^\dagger S_2^\dagger \overline{\chi}_{H2,a_1,a_2}^\dagger \lambda_{H2}^\dagger}{\epsilon} \\ & - h \overline{y}_{2,i_1,r_1} H_{a_2} S_2 y_{2,i_1,r_1} H_{a_1}^\dagger S_2^\dagger \overline{\chi}_{H2,a_1,a_2}^\dagger \lambda_{H2}^\dagger \end{aligned}$ | $\begin{aligned} & + \frac{1}{4} \sqrt{3} h \log \left( \frac{\mu \text{bar}^2}{M_F^2} \right) \overline{y}_{1,i_3,r_1} \overline{L}_{a_1,i_2} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1} \\ & + \frac{1}{4} h \log \left( \frac{\mu \text{bar}^2}{M_F^2} \right) \overline{y}_{2,i_3,r_1} \overline{L}_{a_1,i_2} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1} \\ & + \frac{1}{4} \sqrt{3} h \log \left( \frac{\mu \text{bar}^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} Y_{e,i_3,i_2} \overline{e}_{i_2} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger \\ & + \frac{1}{4} h \log \left( \frac{\mu \text{bar}^2}{M_F^2} \right) \overline{y}_{2,i_1,r_1} Y_{e,i_3,i_2} \overline{e}_{i_2} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger \end{aligned}$ |

[illegible]

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$         | $S_2 F$ | $S_2 H$ | $F H e$                                                                                                                                                                    | $S_1 F H$ |
|-------|----------|-----------|-----------|-----------|-----------|-------|---------------|---------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| T3-E  | 2        | 3         | 1         | 3         | 2         | 2     | $\frac{3}{2}$ |         |         | $+ \frac{1}{4} \sqrt{3} h \log \left( \frac{\mu \text{bar}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}$         |           |
|       |          |           |           |           |           |       |               |         |         | $+ \frac{1}{4} \sqrt{3} h \log \left( \frac{\mu \text{bar}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}$         |           |
|       |          |           |           |           |           |       |               |         |         | $+ \frac{1}{4} \sqrt{3} h \log \left( \frac{\mu \text{bar}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger$ |           |
|       |          |           |           |           |           |       |               |         |         | $+ \frac{1}{4} \sqrt{3} h \log \left( \frac{\mu \text{bar}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger$ |           |

# Field combinations 6 of 6

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | $S_2$                                                                                                                                                                                                             | $F$ | $H$ |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ |                                                                                                                                                                                                                   |     |     |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              |                                                                                                                                                                                                                   |     |     |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  |                                                                                                                                                                                                                   |     |     |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              |                                                                                                                                                                                                                   |     |     |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  |                                                                                                                                                                                                                   |     |     |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ |                                                                                                                                                                                                                   |     |     |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              |                                                                                                                                                                                                                   |     |     |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  |                                                                                                                                                                                                                   |     |     |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              |                                                                                                                                                                                                                   |     |     |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  |                                                                                                                                                                                                                   |     |     |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ |                                                                                                                                                                                                                   |     |     |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              |                                                                                                                                                                                                                   |     |     |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  |                                                                                                                                                                                                                   |     |     |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              |                                                                                                                                                                                                                   |     |     |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  |                                                                                                                                                                                                                   |     |     |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{h \log \left( \frac{\mu h u^2}{M_F^2} \right) \overline{y_{1,i_1,\tau_1}} H_{a_1} H_{a_2} S_{1,j_1} y_{1,i_1,\tau_1} \overline{x_{\tau 3,a_1,a_2,j_1}} \lambda_{\tau 3} S_2^\dagger}{\sqrt{3}}$           |     |     |
|       |          |           |                |           |               |       |                | $-h \log \left( \frac{\mu h a r^2}{M_F^2} \right) \overline{y_{2,i_1,\tau_1}} H_{a_1} H_{a_2} S_{1,j_1} y_{2,i_1,\tau_1} \overline{x_{\tau 3,a_1,a_2,j_1}} \lambda_{\tau 3} S_2^\dagger$                          |     |     |
|       |          |           |                |           |               |       |                | $-2h \log \left( \frac{\mu h a r^2}{M_F^2} \right) \overline{y_{2,i_1,\tau_1}} \overline{y_{e,i_2,i_3}} H_{a_1} S_2 y_{2,i_2,\tau_1} y_{e,i_1,i_3} H_{a_1}^\dagger S_2^\dagger$                                   |     |     |
|       |          |           |                |           |               |       |                | $h \log \left( \frac{\mu h u^2}{M_F^2} \right) \overline{y_{1,i_1,\tau_1}} \lambda_{\tau 3} S_2 y_{1,i_1,\tau_1} \text{Bar}(\text{T3 Mix CG})_{a_1,a_2,j_1} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$    |     |     |
|       |          |           |                |           |               |       |                | $-h \log \left( \frac{\mu h a r^2}{M_F^2} \right) \overline{y_{2,i_1,\tau_1}} \lambda_{\tau 3} S_2 y_{2,i_1,\tau_1} \text{Bar}(\text{T3 Mix CG})_{a_1,a_2,j_1} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$ |     |     |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $-2h \log \left( \frac{\mu h a r^2}{M_F^2} \right) \overline{y_{2,i_1,\tau_1}} H_{a_2} S_2 y_{2,i_1,\tau_1} H_{a_1}^\dagger S_2^\dagger \mathcal{I}_{a a_2,a_1,a_2}^{-1} \lambda_{a a_2}^{-1}$                    |     |     |
|       |          |           |                |           |               |       |                | $-\frac{h \log \left( \frac{\mu h u^2}{M_F^2} \right) \overline{y_{1,i_1,\tau_1}} H_{a_1} H_{a_2} S_{1,j_1} y_{1,i_1,\tau_1} \overline{x_{\tau 3,a_1,a_2,j_1}} \lambda_{\tau 3} S_2^\dagger}{\sqrt{3}}$           |     |     |
|       |          |           |                |           |               |       |                | $-h \log \left( \frac{\mu h a r^2}{M_F^2} \right) \overline{y_{2,i_1,\tau_1}} H_{a_1} H_{a_2} S_{1,j_1} y_{2,i_1,\tau_1} \overline{x_{\tau 3,a_1,a_2,j_1}} \lambda_{\tau 3} S_2^\dagger$                          |     |     |
|       |          |           |                |           |               |       |                | $-2h \log \left( \frac{\mu h a r^2}{M_F^2} \right) \overline{y_{2,i_1,\tau_1}} \overline{y_{e,i_2,i_3}} H_{a_1} S_2 y_{2,i_2,\tau_1} y_{e,i_1,i_3} H_{a_1}^\dagger S_2^\dagger$                                   |     |     |
|       |          |           |                |           |               |       |                | $h \log \left( \frac{\mu h u^2}{M_F^2} \right) \overline{y_{1,i_1,\tau_1}} \lambda_{\tau 3} S_2 y_{1,i_1,\tau_1} \text{Bar}(\text{T3 Mix CG})_{a_1,a_2,j_1} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$    |     |     |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $-h \log \left( \frac{\mu h a r^2}{M_F^2} \right) \overline{y_{2,i_1,\tau_1}} H_{a_2} S_2 y_{2,i_1,\tau_1} H_{a_1}^\dagger S_2^\dagger \mathcal{I}_{a a_2,a_1,a_2}^{-1} \lambda_{a a_2}^{-1}$                     |     |     |
|       |          |           |                |           |               |       |                | $-\frac{h \log \left( \frac{\mu h u^2}{M_F^2} \right) \overline{y_{1,i_1,\tau_1}} H_{a_1} H_{a_2} S_{1,j_1} y_{1,i_1,\tau_1} \overline{x_{\tau 3,a_1,a_2,j_1}} \lambda_{\tau 3} S_2^\dagger}{\sqrt{3}}$           |     |     |
|       |          |           |                |           |               |       |                | $-h \log \left( \frac{\mu h a r^2}{M_F^2} \right) \overline{y_{2,i_1,\tau_1}} H_{a_1} H_{a_2} S_{1,j_1} y_{2,i_1,\tau_1} \overline{x_{\tau 3,a_1,a_2,j_1}} \lambda_{\tau 3} S_2^\dagger$                          |     |     |
|       |          |           |                |           |               |       |                | $-2h \log \left( \frac{\mu h a r^2}{M_F^2} \right) \overline{y_{2,i_1,\tau_1}} \overline{y_{e,i_2,i_3}} H_{a_1} S_2 y_{2,i_2,\tau_1} y_{e,i_1,i_3} H_{a_1}^\dagger S_2^\dagger$                                   |     |     |
|       |          |           |                |           |               |       |                | $h \log \left( \frac{\mu h u^2}{M_F^2} \right) \overline{y_{1,i_1,\tau_1}} \lambda_{\tau 3} S_2 y_{1,i_1,\tau_1} \text{Bar}(\text{T3 Mix CG})_{a_1,a_2,j_1} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$    |     |     |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $-h \log \left( \frac{\mu h a r^2}{M_F^2} \right) \overline{y_{2,i_1,\tau_1}} H_{a_2} S_2 y_{2,i_1,\tau_1} H_{a_1}^\dagger S_2^\dagger \mathcal{I}_{a a_2,a_1,a_2}^{-1} \lambda_{a a_2}^{-1}$                     |     |     |
|       |          |           |                |           |               |       |                | $-\frac{h \log \left( \frac{\mu h u^2}{M_F^2} \right) \overline{y_{1,i_1,\tau_1}} H_{a_1} H_{a_2} S_{1,j_1} y_{1,i_1,\tau_1} \overline{x_{\tau 3,a_1,a_2,j_1}} \lambda_{\tau 3} S_2^\dagger}{\sqrt{3}}$           |     |     |
|       |          |           |                |           |               |       |                | $-h \log \left( \frac{\mu h a r^2}{M_F^2} \right) \overline{y_{2,i_1,\tau_1}} H_{a_1} H_{a_2} S_{1,j_1} y_{2,i_1,\tau_1} \overline{x_{\tau 3,a_1,a_2,j_1}} \lambda_{\tau 3} S_2^\dagger$                          |     |     |
|       |          |           |                |           |               |       |                | $-2h \log \left( \frac{\mu h a r^2}{M_F^2} \right) \overline{y_{2,i_1,\tau_1}} \overline{y_{e,i_2,i_3}} H_{a_1} S_2 y_{2,i_2,\tau_1} y_{e,i_1,i_3} H_{a_1}^\dagger S_2^\dagger$                                   |     |     |
|       |          |           |                |           |               |       |                | $h \log \left( \frac{\mu h u^2}{M_F^2} \right) \overline{y_{1,i_1,\tau_1}} \lambda_{\tau 3} S_2 y_{1,i_1,\tau_1} \text{Bar}(\text{T3 Mix CG})_{a_1,a_2,j_1} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$    |     |     |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $-h \log \left( \frac{\mu h a r^2}{M_F^2} \right) \overline{y_{2,i_1,\tau_1}} H_{a_2} S_2 y_{2,i_1,\tau_1} H_{a_1}^\dagger S_2^\dagger \mathcal{I}_{a a_2,a_1,a_2}^{-1} \lambda_{a a_2}^{-1}$                     |     |     |
|       |          |           |                |           |               |       |                | $-\frac{h \log \left( \frac{\mu h u^2}{M_F^2} \right) \overline{y_{1,i_1,\tau_1}} H_{a_1} H_{a_2} S_{1,j_1} y_{1,i_1,\tau_1} \overline{x_{\tau 3,a_1,a_2,j_1}} \lambda_{\tau 3} S_2^\dagger}{\sqrt{3}}$           |     |     |
|       |          |           |                |           |               |       |                | $-h \log \left( \frac{\mu h a r^2}{M_F^2} \right) \overline{y_{2,i_1,\tau_1}} H_{a_1} H_{a_2} S_{1,j_1} y_{2,i_1,\tau_1} \overline{x_{\tau 3,a_1,a_2,j_1}} \lambda_{\tau 3} S_2^\dagger$                          |     |     |
|       |          |           |                |           |               |       |                | $-2h \log \left( \frac{\mu h a r^2}{M_F^2} \right) \overline{y_{2,i_1,\tau_1}} \overline{y_{e,i_2,i_3}} H_{a_1} S_2 y_{2,i_2,\tau_1} y_{e,i_1,i_3} H_{a_1}^\dagger S_2^\dagger$                                   |     |     |
|       |          |           |                |           |               |       |                | $h \log \left( \frac{\mu h u^2}{M_F^2} \right) \overline{y_{1,i_1,\tau_1}} \lambda_{\tau 3} S_2 y_{1,i_1,\tau_1} \text{Bar}(\text{T3 Mix CG})_{a_1,a_2,j_1} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$    |     |     |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | $S_2 F H$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $-\frac{h \log \left( \frac{\mu \hbar \alpha r_1^2}{M_F^2} \right) \overline{y_{1,i_1,\tau_1}} H_{a_1} H_{a_2} S_{1,j_1} y_{1,i_1,\tau_1} \overline{z_{\tau_3,a_1,a_2,j_1}} \lambda_{\tau_3} S_2^\dagger}{\sqrt{3}}$ $-h \log \left( \frac{\mu \hbar \alpha r_2^2}{M_F^2} \right) \overline{y_{2,i_1,\tau_1}} H_{a_1} H_{a_2} S_{1,j_1} y_{2,i_1,\tau_1} \overline{z_{\tau_3,a_1,a_2,j_1}} \lambda_{\tau_3} S_2^\dagger$ $-2h \log \left( \frac{\mu \hbar \alpha r_2^2}{M_F^2} \right) \overline{y_{2,i_1,\tau_1}} \overline{y_{e,i_2,i_2}} H_{a_1} S_2 y_{2,i_2,\tau_1} Y_{e,i_1,i_1} H_{a_1}^\dagger S_2^\dagger$ $-\frac{h \log \left( \frac{\mu \hbar \alpha r_1^2}{M_F^2} \right) \overline{y_{1,i_1,\tau_1}} \lambda_{\tau_3} S_2 y_{1,i_1,\tau_1} \text{Bar}(\text{T3 Mix CG})_{a_1,a_2,j_1} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}}$ $-h \log \left( \frac{\mu \hbar \alpha r_2^2}{M_F^2} \right) \overline{y_{2,i_1,\tau_1}} \lambda_{\tau_3} S_2 y_{2,i_1,\tau_1} \text{Bar}(\text{T3 Mix CG})_{a_1,a_2,j_1} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$ $-2h \log \left( \frac{\mu \hbar \alpha r_2^2}{M_F^2} \right) \overline{y_{2,i_1,\tau_1}} H_{a_2} S_2 y_{2,i_1,\tau_1} H_{a_1}^\dagger S_2^\dagger \overline{z_{\tau_3,a_1,a_2}^{-1}} \lambda_{\tau_3}^{-1}$ |
|       |          |           |                |           |               |       |                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|       |          |           |                |           |               |       |                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|       |          |           |                |           |               |       |                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|       |          |           |                |           |               |       |                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |